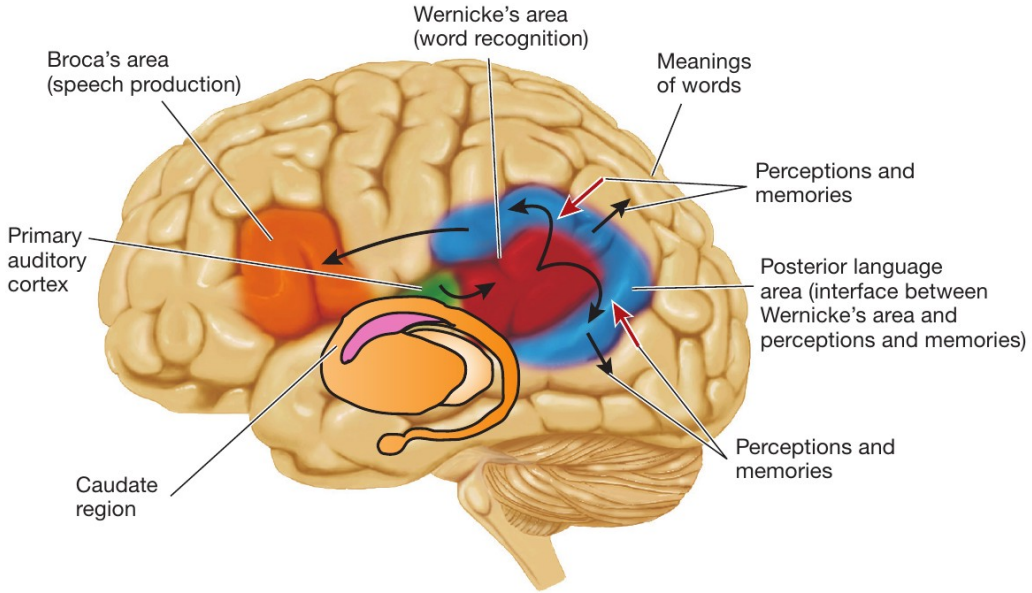


Carlson and Birkett (2022), Fig. 14.3, p. 460.

BEHAVIORAL NEUROSCIENCE 392

Speech, Language, Reading, Writing, and Restored Communication

From Words to Meaning: The Neural Systems of Human Communication

Fall 2026

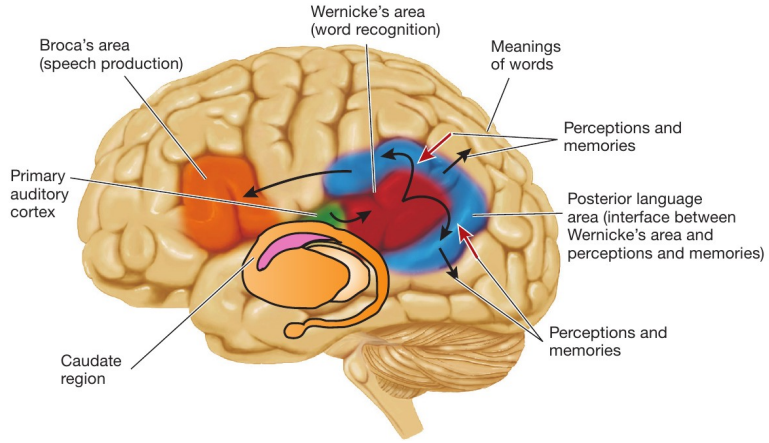
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Language transforms meanings into patterned actions

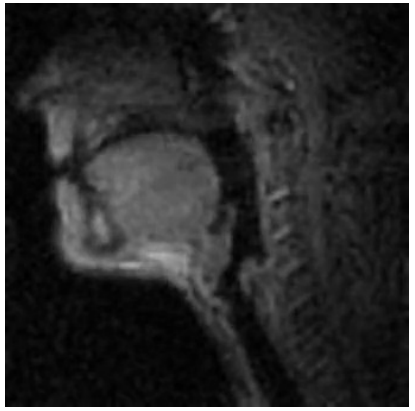
Language allows a person to express ideas through organized symbols. Those symbols can be spoken sounds, written characters, or movements in a signed language. Communication requires a producer to select a message and a receiver to interpret it. The muscles, sensory pathways, and cortical operations involved depend on the form in which that message is expressed.

A speaker selects words, arranges them, and coordinates breathing with movements of the larynx, tongue, lips, and jaw. A listener extracts words from a changing acoustic signal and interprets them using context and stored knowledge. Recognizing the sounds of a word and understanding its meaning are separable operations, although they normally occur together very rapidly.

Brain injury can disrupt one operation while leaving another relatively available. A person might know what an object is but have trouble retrieving its name, or pronounce a written word without understanding it. These dissociations identify component processes. They also explain why the inability to speak does not, by itself, establish a loss of language or intelligence.



Carlson and Birkett (2022), Fig. 14.3, p. 460.



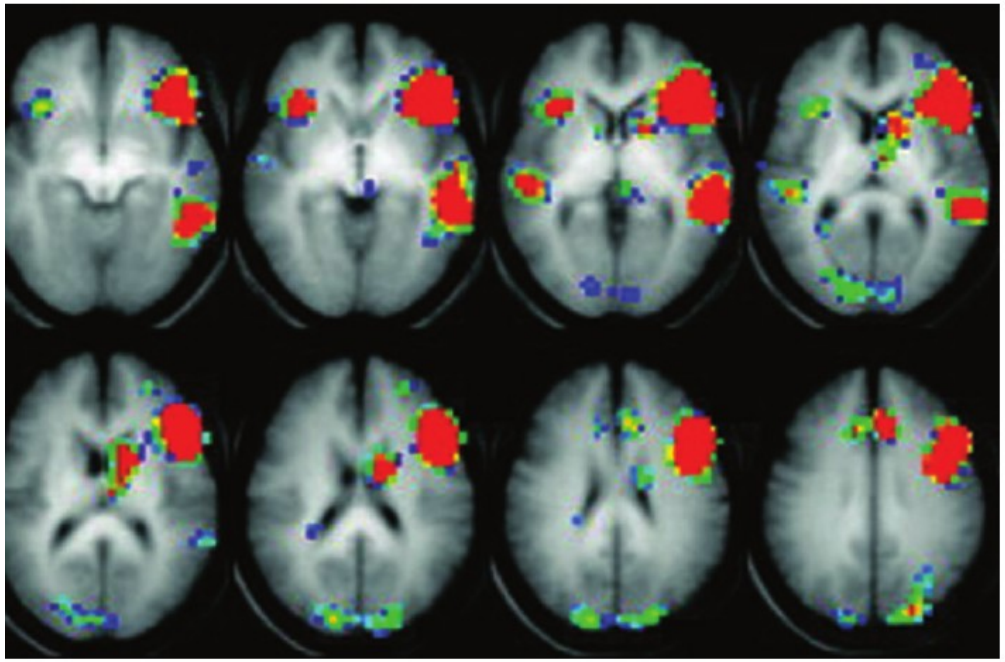
USC SPAN / Esling (2015), rtMRI: rainbow.

Both hemispheres contribute to language

Lateralization means that a function depends more strongly on one cerebral hemisphere than the other. Language production is left dominant in most people, including many people who are left handed. Handedness therefore provides an imperfect prediction of language dominance. Functional imaging and clinical language assessments examine the organization of language more directly in an individual person.

The dominant hemisphere contains interacting frontal, temporal, and parietal regions that contribute to word retrieval, speech organization, and comprehension. Dominance does not mean that the other hemisphere is silent. Both hemispheres process aspects of a conversation, including acoustic information, emotional intonation, context, and the organization of a longer narrative about events.

A lesion produces a deficit according to which tissue and pathways it damages, rather than simply which hemisphere contains it. The same broad diagnostic label can include different combinations of naming, repetition, grammar, and comprehension difficulties. Measuring those components separately gives a clearer description of language function than assuming that one side of the brain performs every linguistic operation.



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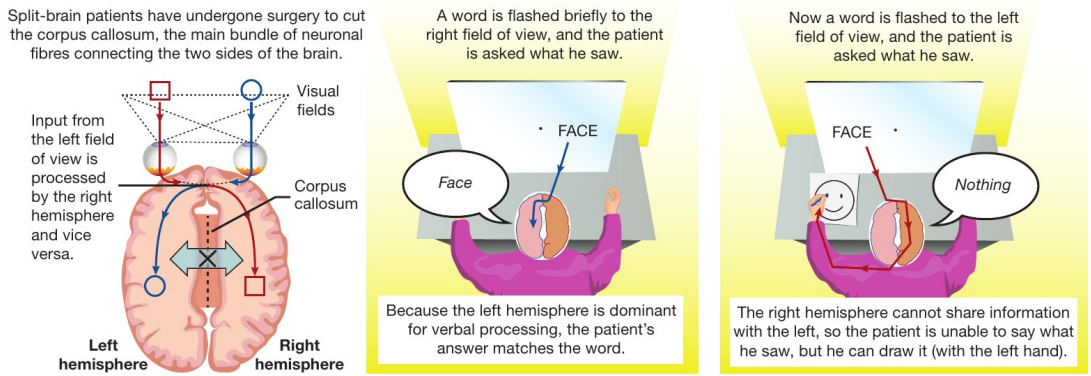
Carlson and Birkett (2022), Fig. 14.1, p. 458.

Callosal disconnection limits verbal access

The corpus callosum is a large bundle of axons that transfers information between the cerebral hemispheres. In some people with severe epilepsy, surgery has divided this pathway to reduce seizure spread. Carefully controlled tests can then present visual information mainly to one hemisphere and examine which responses remain possible when normal interhemispheric exchange is interrupted.

An image briefly presented in the right visual field reaches the left visual cortex. In a person with left dominant speech, that information can usually reach systems that name the image. An image in the left visual field first reaches the right hemisphere. After callosal disconnection, verbal naming of that image can be impaired even when the person selects a matching object with the left hand.

The left hand is controlled mainly by the right hemisphere, so a successful selection can reveal information unavailable to the verbal response system. This result separates perception, access to speech, and the chosen method of reporting. It does not establish that information becomes conscious only when it enters language areas, because awareness cannot be reduced to verbal naming alone.



Carlson and Birkett (2022), Fig. 14.2, p. 459.



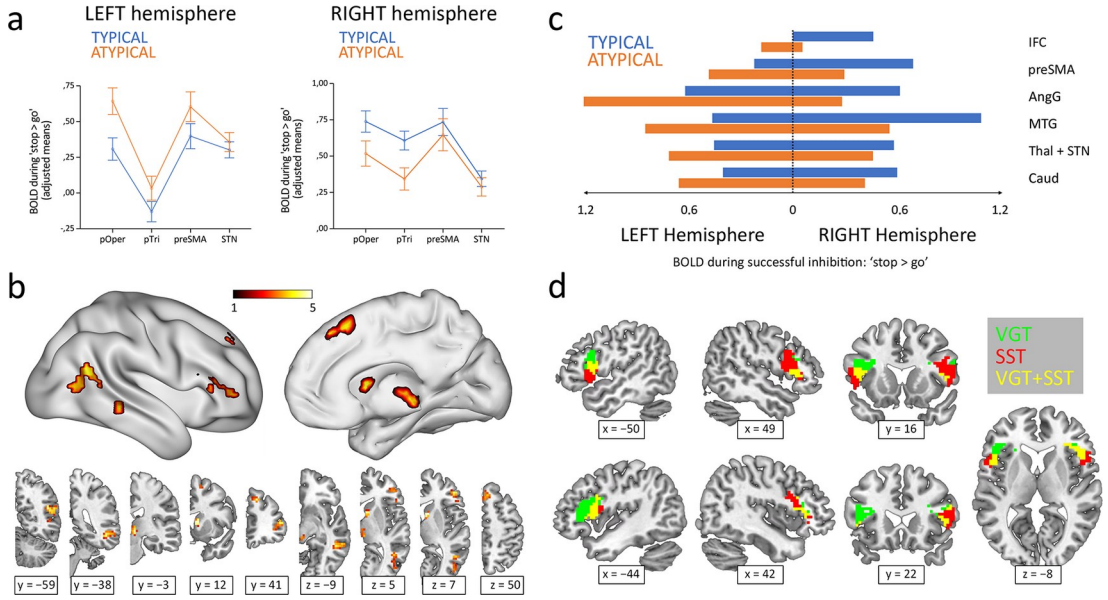
USC SPAN / Esling (2015), rtMRI: grandfather.

Atypical language dominance shifts inhibition

A functional imaging study examined people whose language production was typically left dominant or atypically organized. Participants also performed a stop signal task, which requires canceling a prepared response when a signal appears. This allowed the researchers to compare language lateralization with the organization of a different function that often recruits right inferior frontal cortex.

At the group level, atypical language dominance was accompanied by a shift of the inhibitory control network toward the left hemisphere. The shift included inferior frontal cortex, a medial frontal region involved in preparing actions, and the subthalamic nucleus. A complementary arrangement can therefore occur when language occupies a hemisphere that usually contributes strongly to response inhibition.

The participants did not show a simple division into two perfectly mirrored brain types. Some individuals used overlapping inferior frontal territory for language and inhibitory control. The imaging comparison measured functional organization during two tasks; it did not show that changing language dominance experimentally caused the other network to move to the opposite side of the brain.



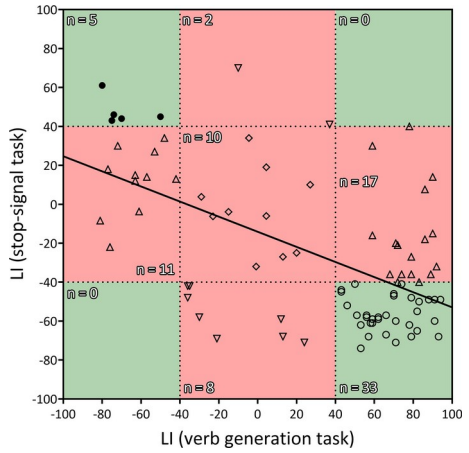
Villar-Rodríguez et al. (2024), Fig. 1.

Frontal language and control regions can overlap

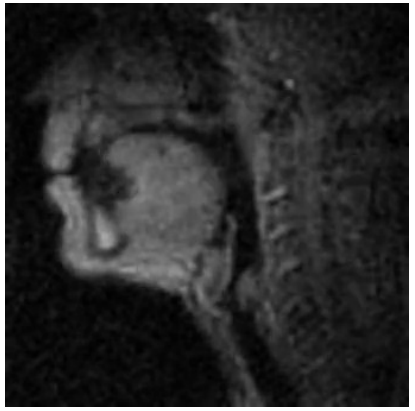
The lateralization study compared each person's language and stop signal measurements rather than relying only on group averages. Individuals with strongly lateralized functions often showed separation across hemispheres. Others showed less segregation and more shared frontal territory. A group map therefore summarizes a distribution of organizations rather than specifying a universal arrangement for every participant.

People with atypical language dominance also had stronger functional communication between hemispheres and a larger corpus callosum in the reported comparisons. The callosum provides anatomical routes for information transfer, while functional connectivity describes coordinated activity. These measurements address different properties and should not be treated as two interchangeable ways of counting the same connections.

Stop signal performance did not differ simply according to whether language was left or right dominant. Successful behavior can therefore arise from more than one neural organization. For clinical assessment, the person's own language and motor maps remain relevant because an unusual distribution of function cannot be predicted accurately from a standard diagram of the dominant hemisphere.



Villar-Rodríguez et al. (2024), Fig. 2.



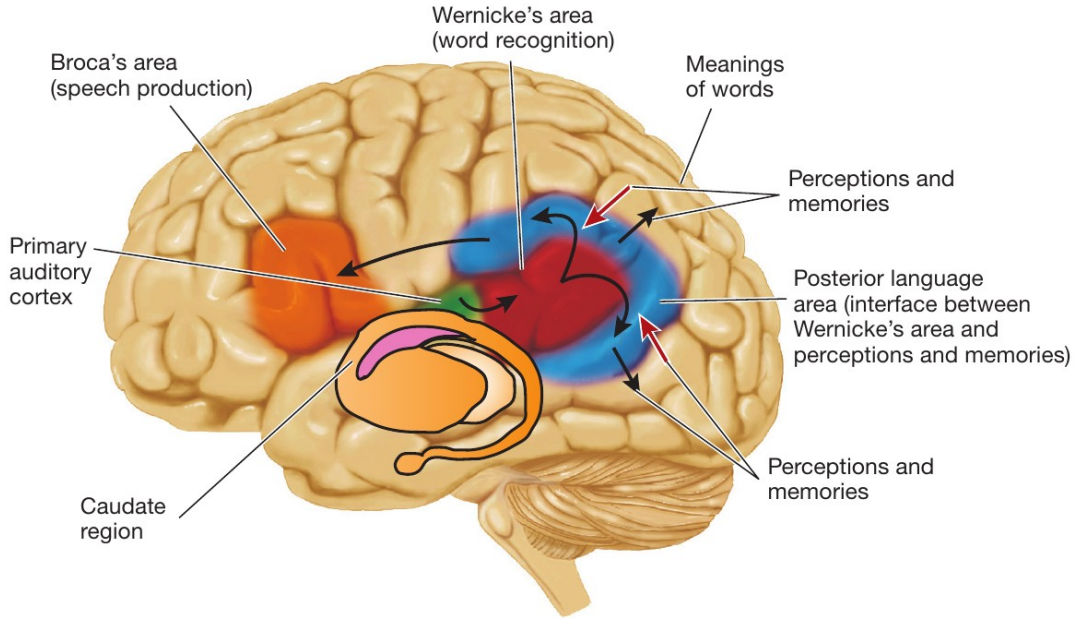
USC SPAN / Esling (2015), rtMRI: your good pants.

Language uses interacting pathways

Language comprehension requires sensory information to reach representations of words and their meanings. Temporal regions analyze features of speech, while additional temporal, parietal, and frontal regions contribute to interpretation and selection. Production also uses these distributed representations, but it must convert a selected message into a sequence of articulatory or manual actions.

A dorsal pathway helps transform speech sounds into movement related representations and supports operations such as repeating an unfamiliar sequence. Ventral pathways contribute to interpreting words and sentences through their meanings. These pathways interact, and neither is a single cable carrying all linguistic information. Damage to their cortical endpoints or connecting white matter can produce different language difficulties.

The familiar Broca and Wernicke diagram identifies important historical landmarks, but fluent communication depends on a larger network. Naming an object requires visual recognition and conceptual knowledge as well as access to a spoken label. Repeating a word can sometimes remain possible when understanding its meaning is impaired, revealing partly separable routes through that network.



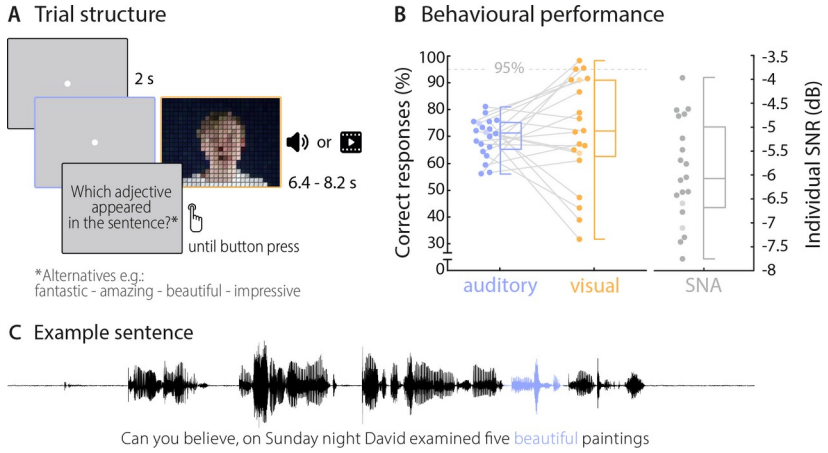
Carlson and Birkett (2022), Fig. 14.3, p. 460.

Speech and lip reading start through different senses

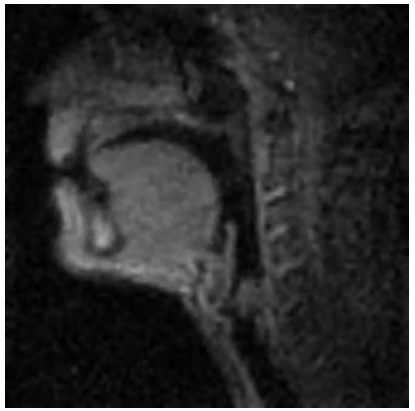
A human magnetoencephalography study compared understanding spoken sentences through sound with understanding silent videos of a speaker's face. Participants answered questions about particular words in the sentences. Magnetoencephalography measures magnetic fields generated by coordinated neural currents, allowing the researchers to examine the timing and location of information about which word had been presented.

The experiment asked two related questions: where neural activity distinguished word identities, and where that activity predicted what the participant understood. A region can encode a feature of the stimulus without providing the most direct explanation of the eventual answer. Separating these questions helped distinguish sensory representation from neural information relevant to successful word comprehension.

In the auditory condition, the changing sound provides cues to consonants and vowels. In the visual condition, mouth movements provide some articulatory cues while leaving other distinctions ambiguous. The study used the same general question structure across conditions, so the comparison concerned different sensory routes into word understanding rather than different definitions of a correct response.



Keitel et al. (2020), Fig. 1.



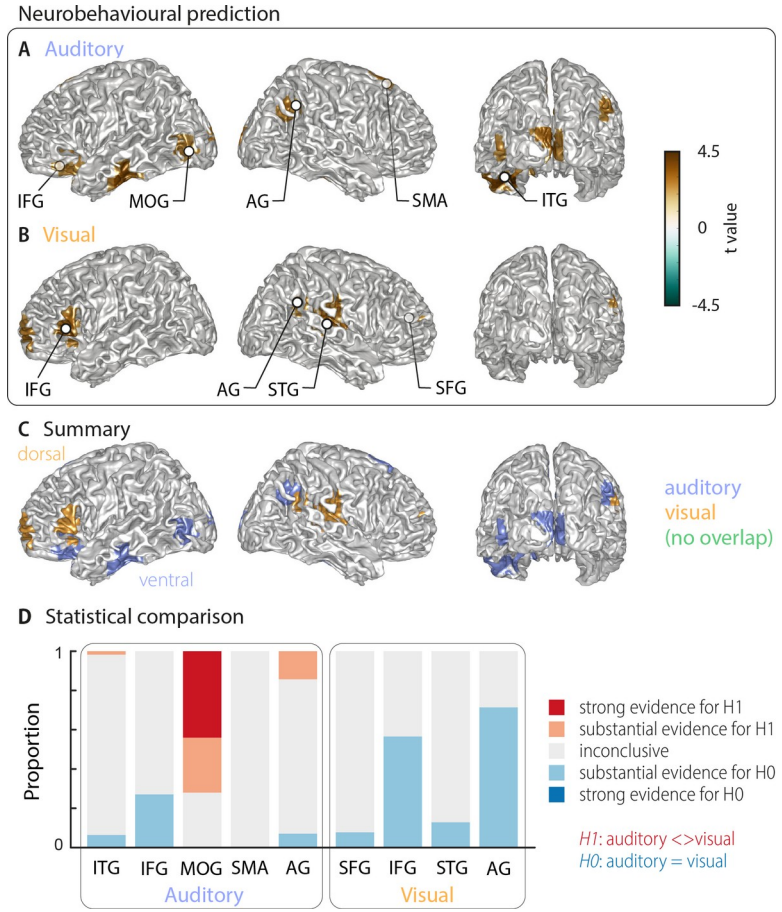
USC SPAN / Esling (2015), rtMRI: the girl was thirsty.

Comprehension signals converge selectively

The auditory and visual speech study found largely distinct sensory driven representations of word identity. Neural representations that predicted the participant's answer converged more selectively in anterior angular and inferior frontal regions. Shared comprehension therefore appeared at particular stages of processing, rather than as one completely identical neural representation across every brain area and sensory condition.

The angular gyrus lies in the inferior parietal lobe, and inferior frontal cortex contributes to selecting and organizing language related information. Their participation was identified from relationships between neural patterns and trial by trial understanding. A region's average activation alone would not reveal whether its activity distinguished the word that the person actually reported.

Lip reading supplies incomplete information because different speech sounds can produce similar visible mouth shapes. Context and stored language knowledge help resolve that ambiguity. The findings explain how modality specific sensory signals can contribute to a shared interpretation while preserving differences in the neural representations through which auditory and visual speech enter the language system.



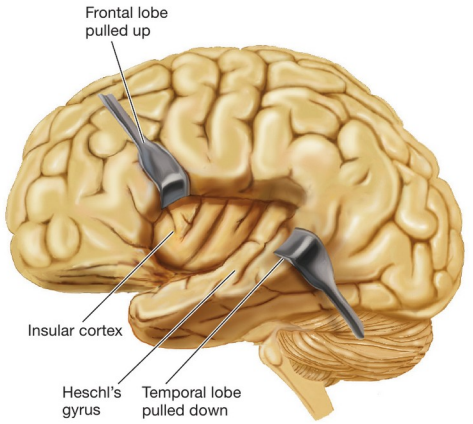
Keitel et al. (2020), Fig. 3.

Speech requires coordinated articulatory movement

Articulation is the movement of the vocal tract that shapes speech sounds. Air from the lungs passes through the larynx, and movements of the tongue, lips, jaw, and soft palate alter the resulting sound. A consonant may require a brief closure, a narrow constriction, or a coordinated change in voicing as air moves through the tract.

The brain must organize these movements over time because neighboring sounds influence one another. The tongue position for a consonant can change with the vowel that follows it. This overlap is called coarticulation. It allows rapid, continuous speech but means that spoken words cannot be produced by simply playing a fixed independent motor command for each letter.

Motor and premotor cortices contribute to preparing and executing articulatory movements, while sensory feedback provides information about their consequences. Auditory feedback reports the produced sound, and somatosensory feedback reports contact and movement within the vocal tract. Language formulation and motor execution can fail separately, which distinguishes aphasia from a speech difficulty caused primarily by impaired movement.



Carlson and Birkett (2022), Fig. 14.9, p. 467.



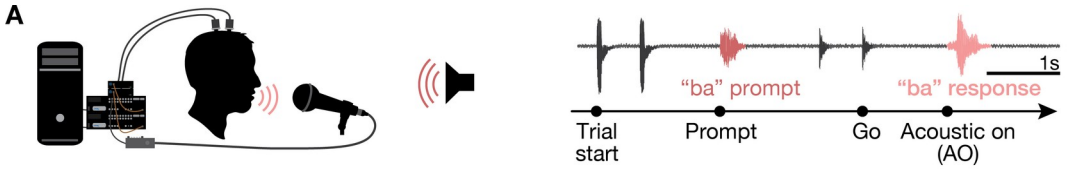
USC SPAN / Esling (2015), rtMRI: bilabial plosive voiceless.

Dorsal motor cortex also responds during speech

An intracortical recording study examined two people with tetraplegia who had electrode arrays in the dorsal motor cortex hand region. They heard a word or syllable and then spoke after a go cue. The arrays recorded neural spiking and related signals at a much finer spatial scale than conventional functional imaging can usually provide.

Neurons in this region changed their firing during speaking and during nonspeech movements of the tongue, lips, and jaw. This was unexpected from a strict interpretation of the motor homunculus as sharply separated body part compartments. At the single neuron and population levels, motor representations can extend beyond the simplified boundaries shown in a standard cortical map.

The participants could produce speech, so the study measured actual speaking rather than reconstructed speech in people who could not vocalize. Simultaneous audio identified when articulation occurred. Comparing neural activity around the prompt, go cue, and acoustic onset separated responses associated with hearing an instruction from activity associated with initiating and producing the spoken response.



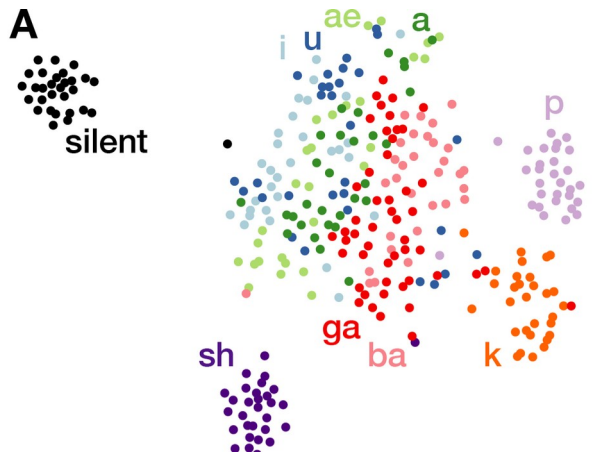
Stavisky et al. (2019), Fig. 1.

Neural patterns distinguish speech items

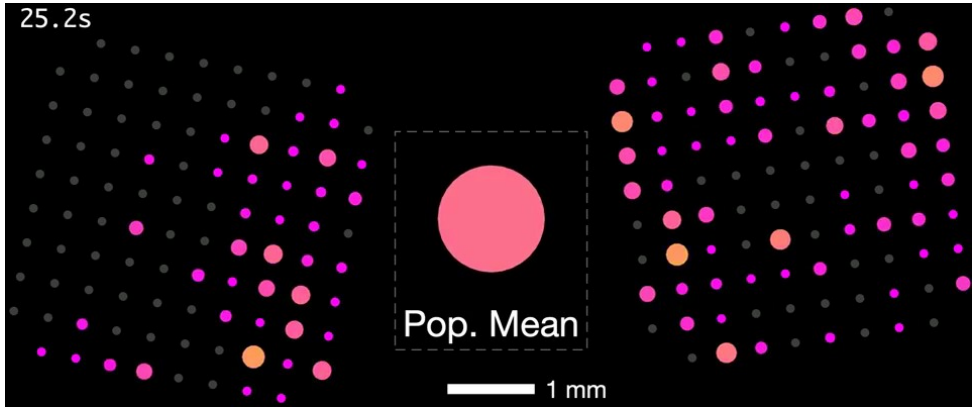
The dorsal motor cortex study used patterns across electrodes to classify which syllable or word a participant produced. A decoder is a computational procedure that maps measured activity onto an estimated output. Here, neural signals contained enough information to distinguish speech items on individual trials rather than only after averaging many repetitions of the same utterance.

The researchers also examined population dynamics, meaning how the combined pattern of activity changed over time. A broadly shared initiation component preceded more item dependent changes during speaking. Rotational trajectories appeared in a reduced representation of that activity, resembling dynamical motifs previously reported during arm movements in other motor cortex experiments.

A successful classifier establishes that a recorded signal contains useful information about the task, but the classifier's categories are not a literal list of words stored inside each neuron. Several neurons contribute together, and individual neurons can respond to multiple movements. Speech neuroprostheses exploit these distributed patterns while also requiring a separate method for translating them into usable communication.



Stavisky et al. (2019), Fig. 3.



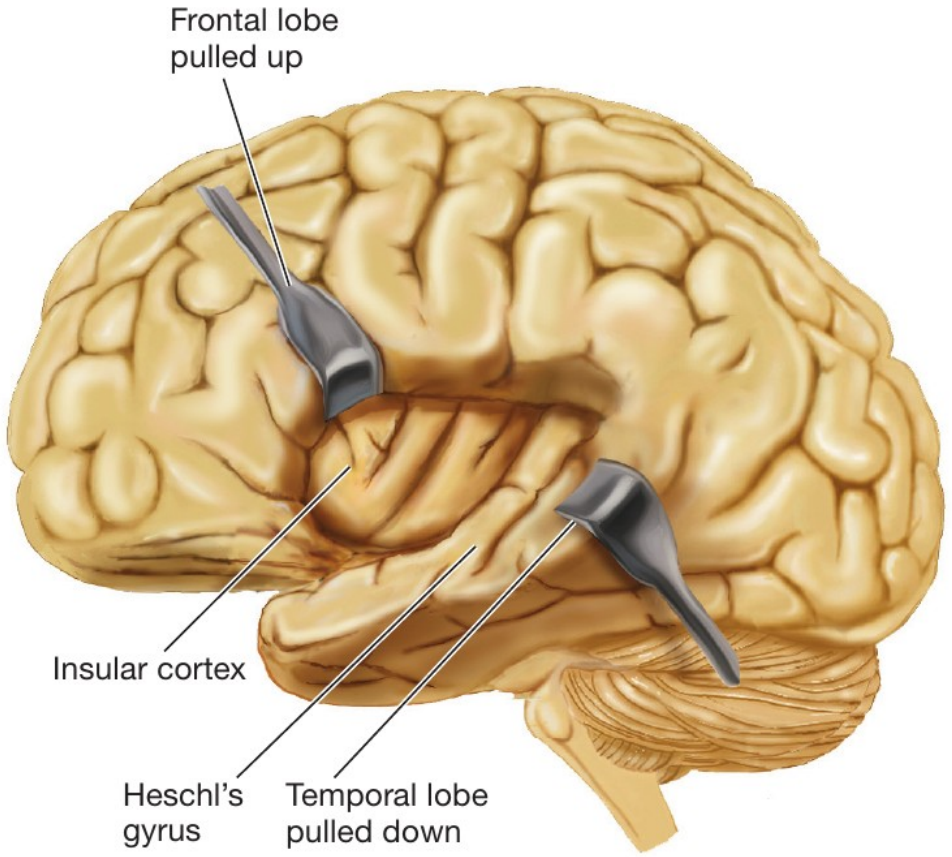
Stavisky et al. (2019), Video 1.

The insula acts within a wider speech network

The insula is cortex folded beneath the lateral surface of the brain, covered by neighboring frontal, parietal, and temporal regions. Lesion studies have associated damage near the insula with difficulty planning speech movements. However, lesions often affect several nearby structures and pathways, making it necessary to distinguish insular activity from activity in the surrounding frontal operculum.

Apraxia of speech refers to impaired planning or programming of speech movements that cannot be explained simply by muscle weakness. A person may make inconsistent sound errors and struggle to sequence articulatory gestures. This differs from difficulty retrieving a word's meaning, although both problems can occur together after a larger lesion in the dominant hemisphere.

Direct intracranial recordings can determine whether a region becomes active before articulation or mainly after movement begins. That temporal distinction matters when evaluating a proposed planning role. A response that follows speech onset may reflect sensory consequences, ongoing coordination, or another process, rather than the preparation that generated the first articulatory movement.



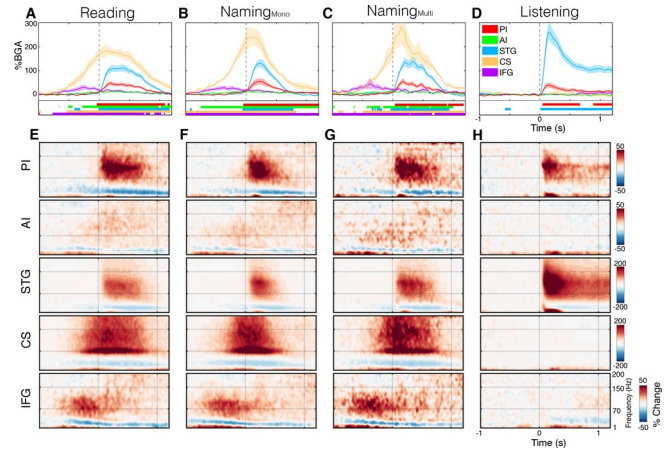
Carlson and Birkett (2022), Fig. 14.9, p. 467.

Posterior insula responds after articulation starts

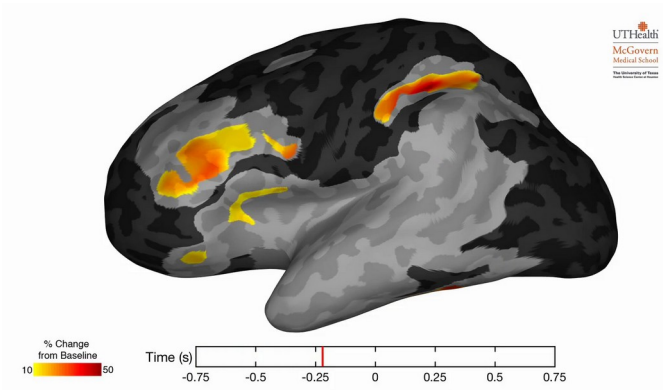
A study using penetrating intracranial electrodes recorded activity during single word production, speech listening, and nonspeech mouth movements. The investigators compared anterior and posterior insular regions with neighboring structures. Words varied in articulatory complexity, allowing the timing of regional activity to be examined across several forms of speaking rather than one isolated utterance.

Posterior insula responded robustly on both sides after articulation began. It also responded during speech listening and nonspeech orofacial movements. In contrast, activity before articulation was sparse and was found mainly in the frontal operculum rather than the insula. The temporal pattern did not match a strong general role for insular cortex in preparing every spoken movement.

Posterior insular activity was stronger for self generated speech than for externally heard speech, whereas superior temporal cortex showed the opposite comparison. This dissociation suggests different contributions to processing a person's own vocal behavior and incoming sounds. The recordings refine the anatomical interpretation of lesion findings without equating every response during speaking with a speech planning command.



Woolnough et al. (2019), Fig. 2.



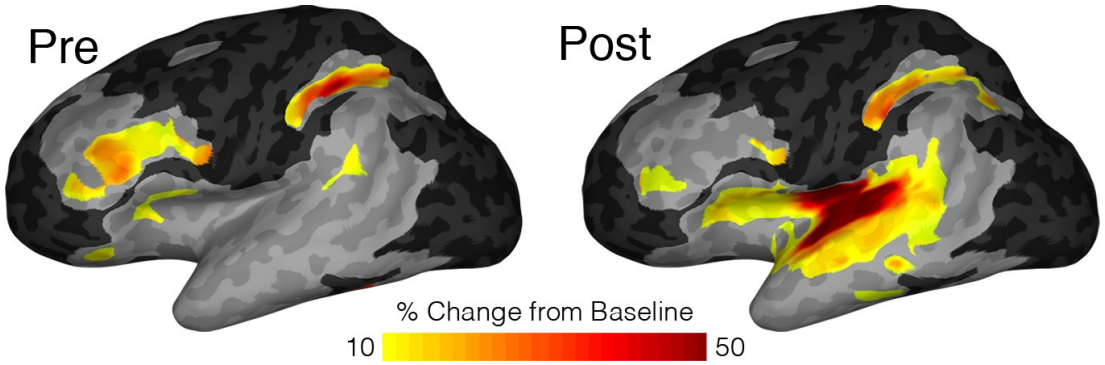
Woolnough et al. (2019), Video 1.

Speech activity changes before and after onset

The intracranial insula study mapped activity before and after the onset of reading aloud. Before articulation, the frontal pattern differed from the later pattern that included posterior insular and temporal regions. The investigators aligned recordings to speech onset so that neural events occurring before movement could be distinguished from responses accompanying its sensory and motor consequences.

The recordings measured broadband high frequency activity, a signal associated with local population activity. This is not a direct count of all action potentials or of neurotransmitter release. Comparing the same signal across recording sites and task periods nevertheless provided a temporally precise description of which sampled regions participated as the utterance unfolded.

Electrode placement was determined by clinical care, so the map included only regions with adequate recording coverage. The published movie identifies excluded territory rather than depicting it as inactive cortex. Reading the spatial map together with its time axis prevents a later response from being mislabeled as the neural event that prepared the spoken word.



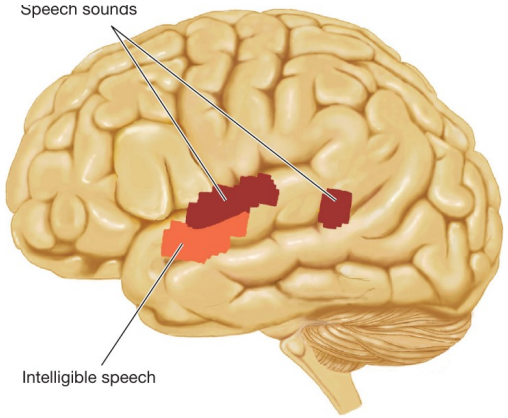
Woolnough et al. (2019), Fig. 4.

Shared cortex can perform different operations

Listening to a syllable and producing that syllable share its linguistic identity but differ in their immediate biological requirements. Listening begins with an external acoustic signal. Speaking requires commands to the vocal tract and produces auditory and somatosensory feedback. A cortical region that participates in both tasks may therefore represent different information in each condition.

High frequency activity in motor cortex can increase during speech listening, but this does not automatically mean that a listener reconstructs the exact movements required to produce every sound. To test that claim, researchers compare the structure of neural response patterns. If the same articulatory representation is reused, sounds with similar movements should produce correspondingly similar patterns during listening.

Acoustic similarity and articulatory similarity are related but distinguishable. For example, place of articulation concerns where airflow is constricted, while acoustic features concern the resulting sound. A study can compare neural pattern similarity with each description. This approach asks what information is represented rather than treating any activation in motor cortex as proof of a particular theory of perception.



Carlson and Birkett (2022), Fig. 14.12, p. 469.



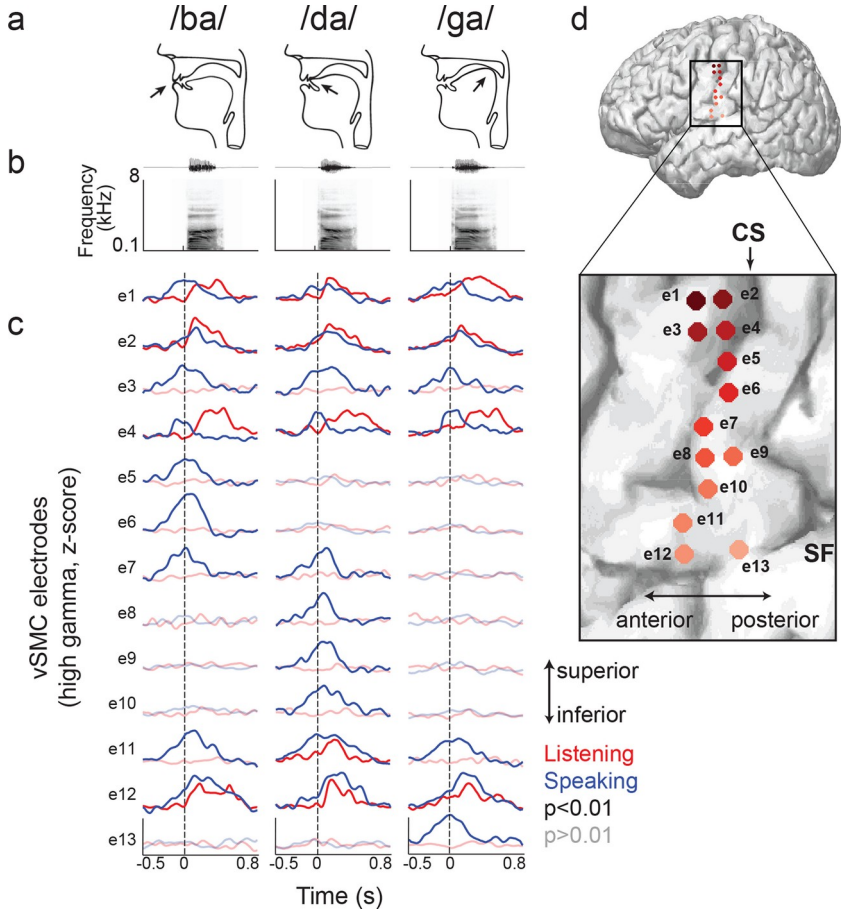
USC SPAN / Esling (2015), rtMRI: alveolar plosive voiceless.

Motor cortex codes heard and spoken syllables differently

A direct cortical recording study compared neural responses while people listened to and articulated consonant vowel syllables. The same speech items allowed acoustic identity to be held comparable across tasks. Electrodes sampled ventral sensorimotor cortex, where movements of speech articulators are represented, and the investigators examined activity at individual sites and across the electrode population.

During speaking, responses extended across representations of speech articulators. During listening, activity appeared mainly in superior and inferior portions of ventral motor cortex. The response at a particular site could therefore differ substantially between hearing and producing the same syllable, even though both events were associated with activity somewhere in the broader motor region.

The different distributions show why the location of an active region cannot by itself identify its computation. Motor cortex receives sensory information as well as issuing movement related signals. During perception, its activity may represent incoming vocal information in a form that differs from the articulatory patterns used when the listener becomes the speaker.



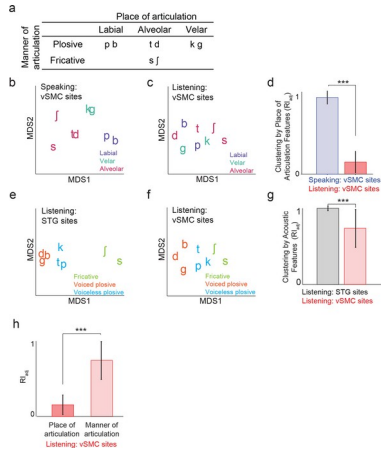
Cheung et al. (2016), Fig. 2.

Heard speech responses emphasize acoustic similarity

The listening and speaking study compared how similar the neural population patterns were for different syllables. During articulation, pattern organization reflected features of the movements used to produce the sounds. During listening, the organization instead resembled acoustic features, more like the arrangement measured in auditory cortex than a replay of the speaker's articulatory map.

This result directly tests a representational prediction rather than merely asking whether motor cortex is active. Two sounds can evoke activity in the same general region while generating different relationships across electrodes. Those relationships reveal whether the sampled population groups sounds by the movements that create them or by properties of the sounds arriving at the ear.

The study does not remove motor systems from communication. Speaking, imitation, and adjusting one's own vocal output still require sensorimotor operations. Its specific finding concerns the code measured during listening: the recorded motor cortical responses carried auditory vocal information, so their presence did not establish that speech perception depends on recreating the corresponding articulatory gestures.



Cheung et al. (2016), Fig. 4.



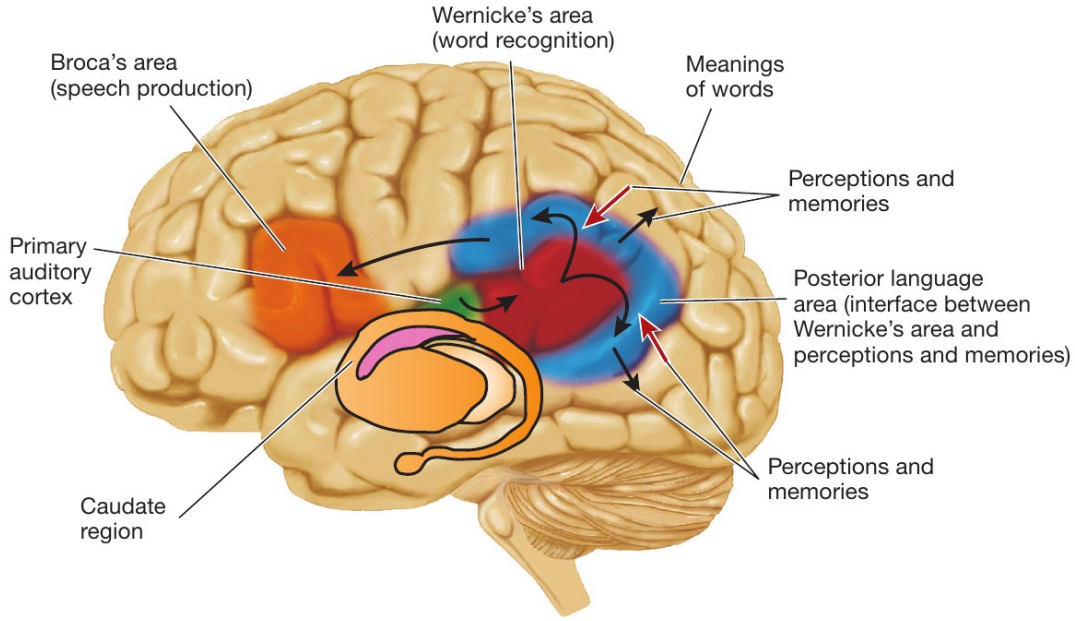
USC SPAN / Esling (2015), rtMRI: velar plosive voiceless.

Word meanings draw on conceptual knowledge

Semantics concerns the meanings of words and other symbols. Understanding the word lemon can involve knowledge of its appearance, taste, typical uses, and category. Those features were acquired through several kinds of experience and are represented across interacting brain systems. A spoken or written word can activate that knowledge without the object itself being present.

Temporal association cortex contributes to representing conceptual relationships, while frontal and parietal regions help retrieve or select information according to the current task. A person deciding whether a lemon is edible uses a different part of their knowledge than a person deciding whether it is usually yellow. The same word can therefore guide different neural and behavioral responses.

The historical brain dictionary diagram summarizes transformations between word forms and meanings, but meanings are not stored as a single verbal definition at one point in cortex. Damage can disturb recognition of a word's sound, access to its concept, or retrieval of its name. Testing several input and output modalities helps determine which transformation has been impaired.



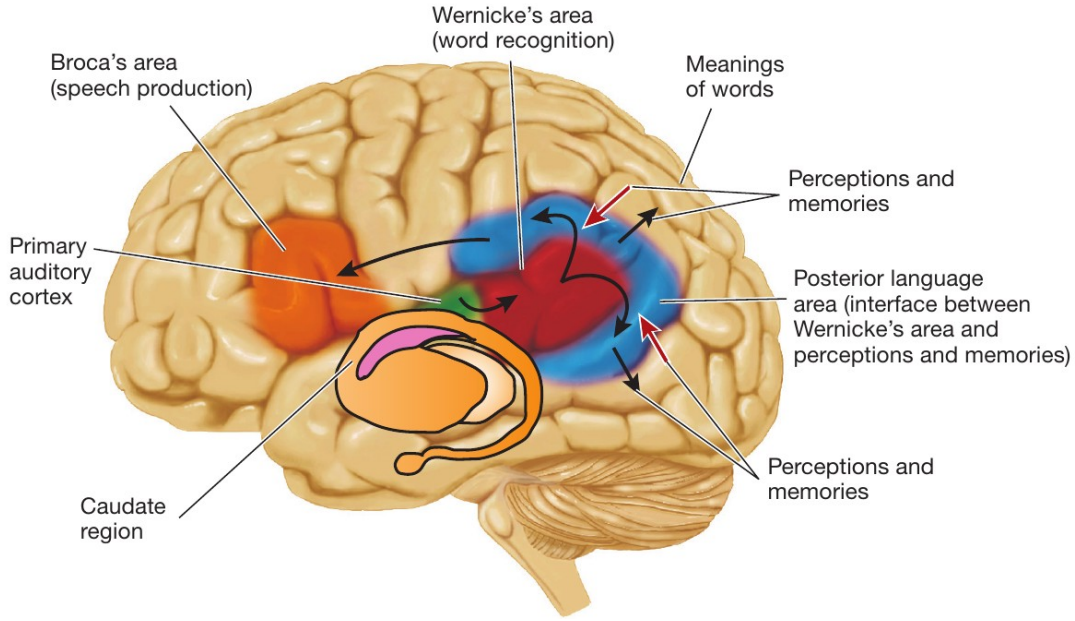
Carlson and Birkett (2022), Fig. 14.3, p. 460.

Syntax organizes relationships among words

Syntax is the organization of words into structured relationships. The sentences the dog chased the child and the child chased the dog use the same words but assign different roles. Comprehension therefore requires more than recognizing each item. The listener must determine who performed an action, who received it, and how clauses depend on one another.

As a sentence unfolds, the brain maintains and revises an interpretation using grammatical cues, word meanings, and context. An early phrase may permit several possible analyses until a later verb resolves the ambiguity. This incremental process allows understanding before the final word arrives, but it can also require revision when the continuation contradicts the initial interpretation.

Frontal and temporal regions contribute to sentence level processing through interacting operations. A grammar difficulty can coexist with relatively preserved understanding of individual concrete words. Conversely, a person can produce a grammatically shaped sentence whose content is inappropriate because word selection or conceptual knowledge is impaired. These patterns separate sentence organization from the meanings of its individual components.



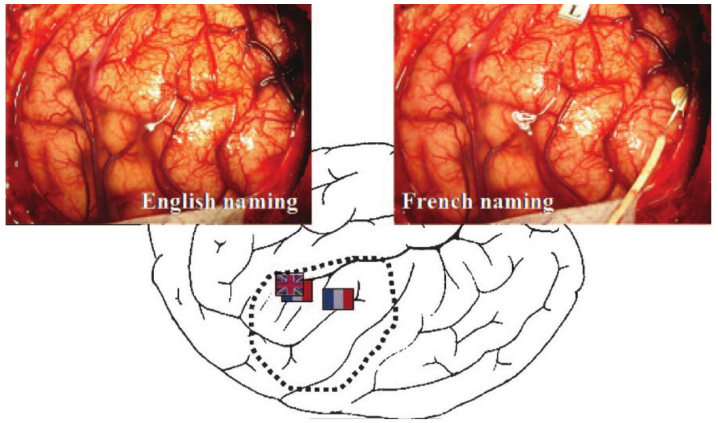
Carlson and Birkett (2022), Fig. 14.3, p. 460.

Bilingual language networks overlap

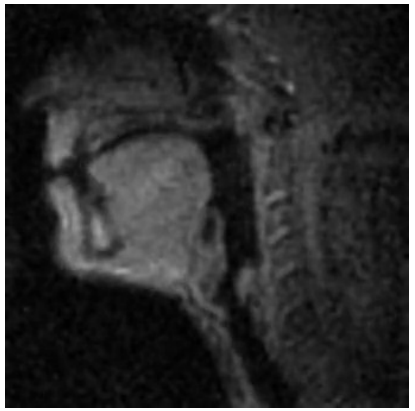
Bilingual people use two languages whose sounds, words, and grammatical patterns may differ. Many language operations recruit overlapping cortical networks, but the amount of overlap depends on the task, proficiency, learning history, and particular language pair. Knowing two languages does not require two completely independent sets of cortical regions devoted exclusively to each language.

Producing a message in one language also requires selecting the appropriate words while alternatives from another language may be available. Frontal and subcortical control systems contribute to selection and switching. A person's ability to retrieve a particular word can differ between languages because frequency of use and contexts of learning differ across the two vocabularies.

Injury can impair languages unequally or affect the ability to keep them separate. Clinical mapping therefore tests the languages that matter to the person rather than assuming that performance in one predicts all others. Language experience also changes expectations about timing and word order, allowing researchers to ask how learned linguistic regularities influence later perception.



Carlson and Birkett (2022), Fig. 14.5, p. 462.



USC SPAN / Esling (2015), rtMRI: she had your dark suit.

Listeners revise sentence interpretations

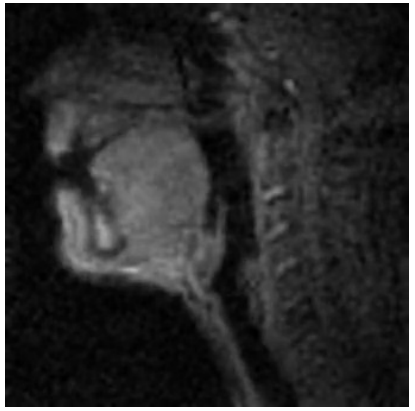
A study of incremental speech comprehension used sentences that could initially support more than one grammatical interpretation. The meaning and grammatical behavior of an early verb influenced which interpretation was plausible. A later main verb then provided information that could favor one structure over another, making the sentence a test of interpretation as it unfolded.

The researchers recorded electrical and magnetic brain signals while participants listened. They also derived word by word structural measures from a language model called BERT. The model supplied a description of how alternative interpretations changed with incoming words; the neural recordings supplied an independent measurement of brain activity during the same sentences.

Interpreting a sentence requires integrating multiple constraints rather than consulting one grammatical rule in isolation. The likely role of a noun, the kind of verb, and the surrounding context all affect the developing message. By varying these features in matched sentences, the study examined how neural processing tracks changes in structured interpretation over time.



Lyu et al. (2024), Fig. 1.



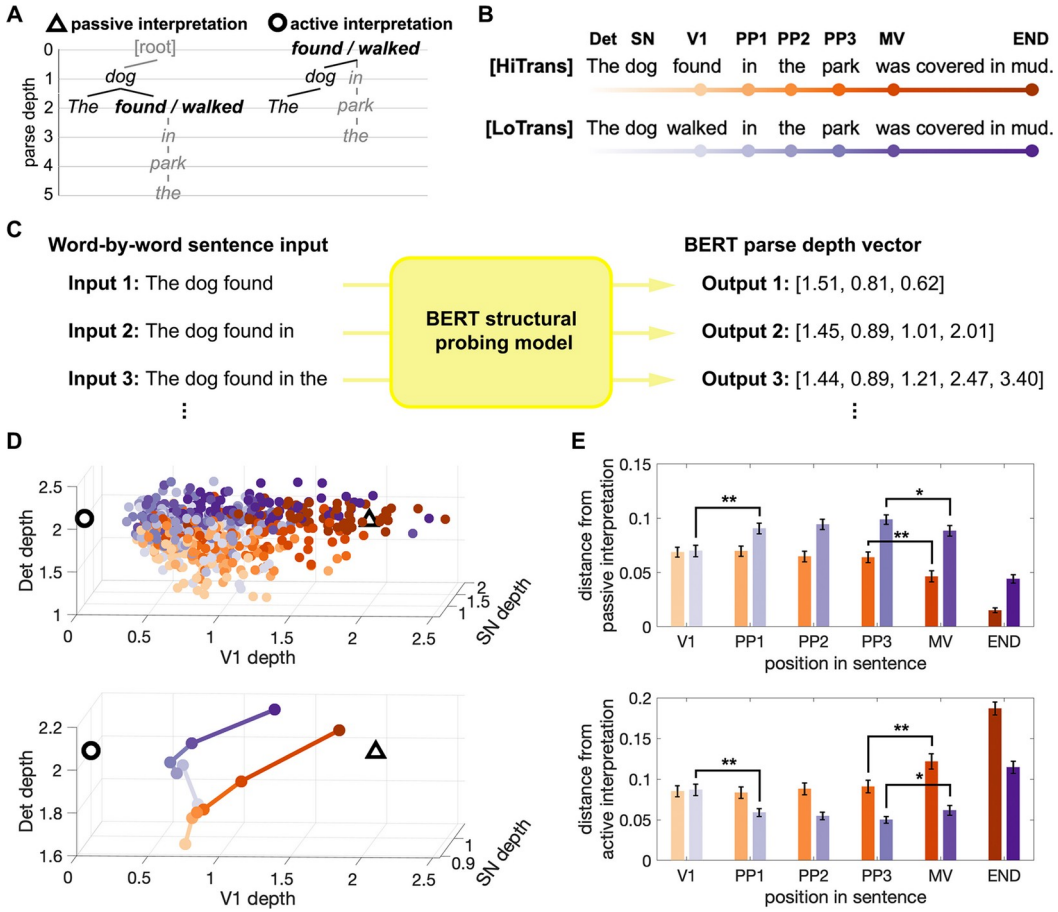
USC SPAN / Esling (2015), rtMRI: dont ask me to carry.

Parse estimates can track neural processing

The sentence comprehension study represented grammatical relationships with dependency trees. In such a tree, words are organized according to which other words they modify or depend on. The researchers asked how a model's estimated structure changed as each new word was added, providing an explicit account of the competing interpretations rather than labeling an entire sentence as simply difficult.

They then compared structural measures and other linguistic information with time resolved brain responses. The findings described a distributed process involving bilateral regions beyond the classical frontal and temporal language landmarks. Different constraints contributed as the sentence unfolded, rather than one region waiting until the end to reconstruct the whole message from an unorganized list of words.

The published model diagram depicts the analysis used to characterize possible interpretations, not a microscope view of a neural circuit. A similarity between a model measure and a brain signal identifies useful explanatory information. It does not establish that human neurons implement the model's artificial architecture or that the model comprehends a sentence in the same way as a person.



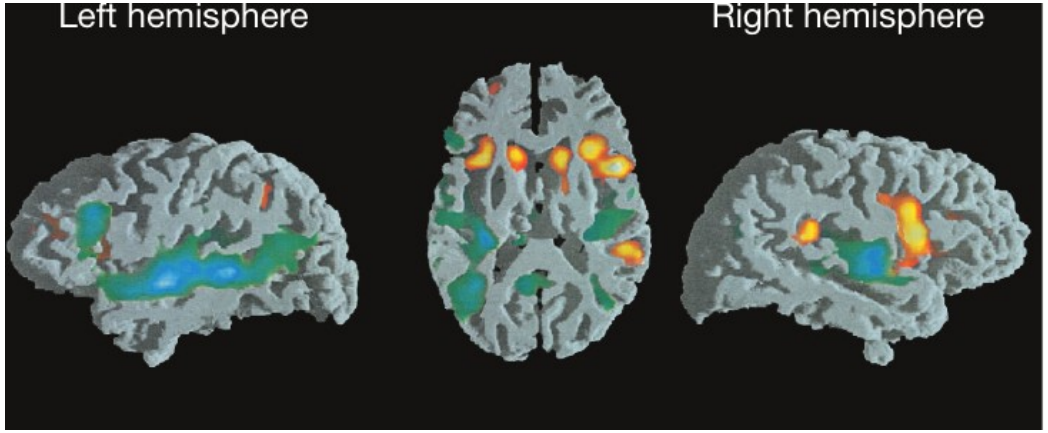
Lyu et al. (2024), Fig. 3.

Prosody supplies rhythm emphasis and tone

Prosody is the pattern of pitch, timing, loudness, and stress across speech. The same sequence of words can sound like a question, a statement, or an expression of disbelief depending on these features. Prosody also marks which parts of a sentence belong together, helping a listener group a continuous acoustic signal into meaningful phrases.

Emotional prosody contributes information about a speaker's state, while linguistic prosody can distinguish word stress or sentence structure. These functions overlap in their acoustic cues but are not identical. A listener can recognize the literal words while having difficulty identifying their emotional tone, showing that access to vocabulary does not guarantee normal interpretation of intonation.

Both hemispheres contribute to prosodic processing, with right temporal and frontal systems often important for emotional tone. The relative contribution depends on the task and the information being extracted. Experience with a language also establishes expectations about duration and stress, so a rhythm that seems natural to one listener can reflect learned regularities rather than a universal acoustic grouping rule.



Carlson and Birkett (2022), Fig. 14.7, p. 463.



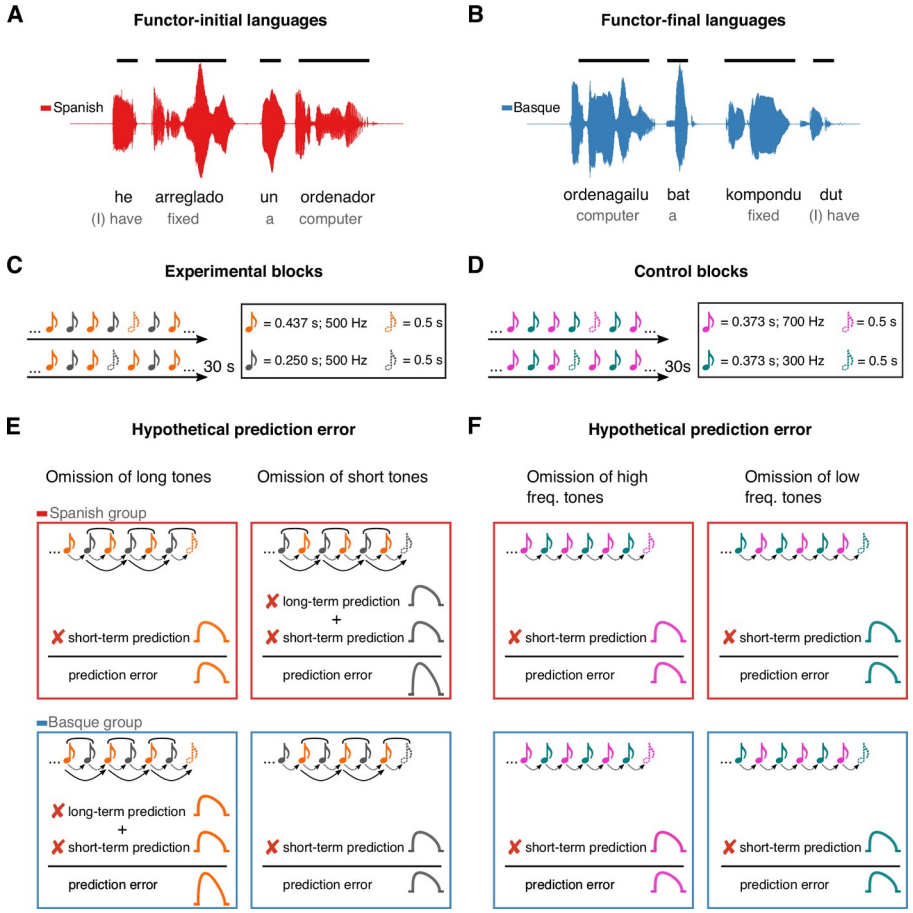
USC SPAN / Esling (2015), rtMRI: glottal plosive.

Language experience shapes timing predictions

A magnetoencephalography study compared native Spanish and Basque speakers listening to repeating sequences of simple tones. These languages differ in typical word order and associated prosodic patterns. The experiment asked whether long term language experience influenced predictions about sound timing even when the stimuli were not spoken words and carried no sentence meaning.

Tones alternated in duration, and occasional omissions violated the expected sequence. The brain's response to a missing sound provides evidence about what had been predicted, because no acoustic event occurs at the omitted position. A control condition alternated pitch instead of duration, allowing the investigators to test whether any difference was specific to the timing cue of interest.

The comparison combined short term regularities established within the tone sequence with longer term experience associated with the native language. The two sources of prediction could therefore be examined separately. Participants were not learning a new grammar during the experiment; they were listening with auditory expectations shaped by their existing linguistic environment.



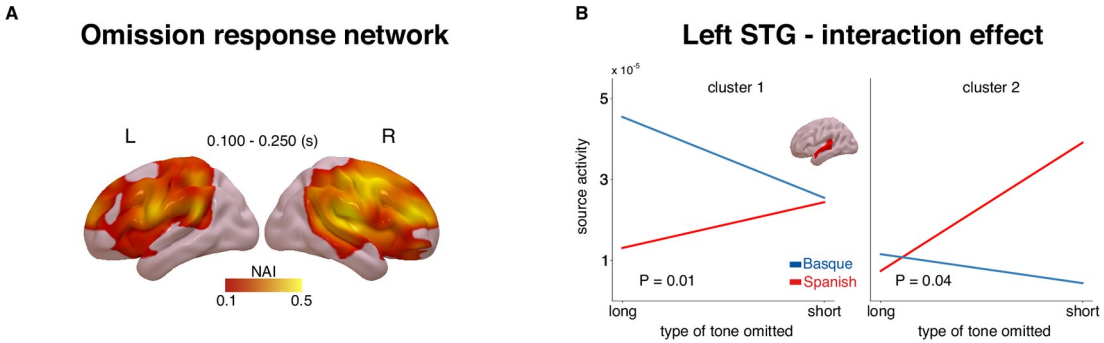
Morucci et al. (2024), Fig. 1.

Omissions reveal learned auditory expectations

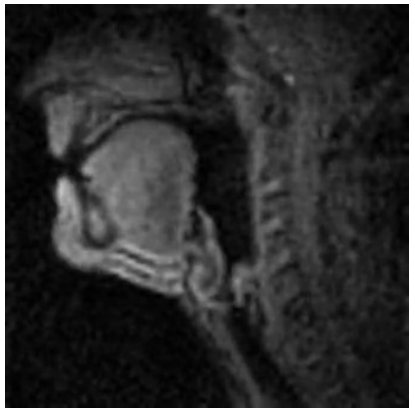
The Spanish and Basque groups differed in neural responses to omitted tones when the sequences alternated in duration. The pattern depended on the listeners' linguistic background and appeared strongly in left auditory cortex. A missing event therefore evoked different responses according to the long term timing expectations that listeners brought into the same acoustic situation.

When the tones alternated in pitch instead, the corresponding group difference was absent. This control narrowed the result to the duration based prediction tested by the study. It did not show that speakers of different languages process every sound differently; the effect concerned a particular learned regularity interacting with the structure of the experimental sequence.

The measured signal was an omission related magnetic response, not a verbal judgment about grammar. Language experience can therefore influence processing at an auditory stage even when people do not consciously translate sounds into words. The experiment connects linguistic history to a specific physiological prediction about when an expected sound should have occurred.



Morucci et al. (2024), Fig. 3.



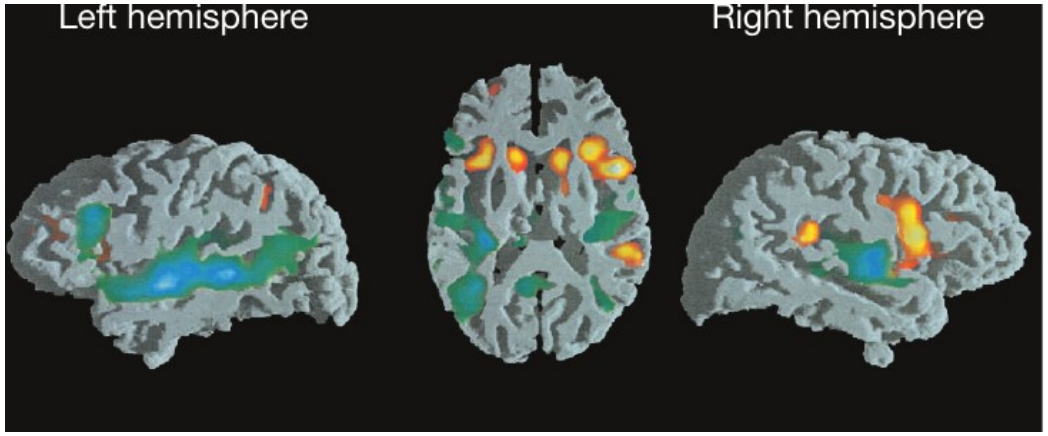
USC SPAN / Esling (2015), rtMRI: heed.

Voice identity differs from linguistic content

A voice carries information about both the message and the speaker. A listener can recognize a familiar person across different words, while also understanding the same word spoken by several unfamiliar people. Voice recognition therefore requires extracting relatively stable identity cues from an acoustic signal that also changes with articulation, emotion, and recording conditions.

Temporal cortical systems contribute to analyzing voice features, and damage can impair recognition of familiar voices without eliminating all speech comprehension. The reverse dissociation can also occur: someone may identify who is speaking while failing to understand the words. These patterns separate the analysis of personal identity from the transformations that recover linguistic meaning.

Emotional tone provides another dimension carried by the voice. Pitch and timing can signal happiness or uncertainty without changing the words themselves. Researchers can compare vocal and matched nonvocal sounds to test whether a response reflects general acoustic differences or sensitivity to the biologically meaningful organization of human vocal expressions.



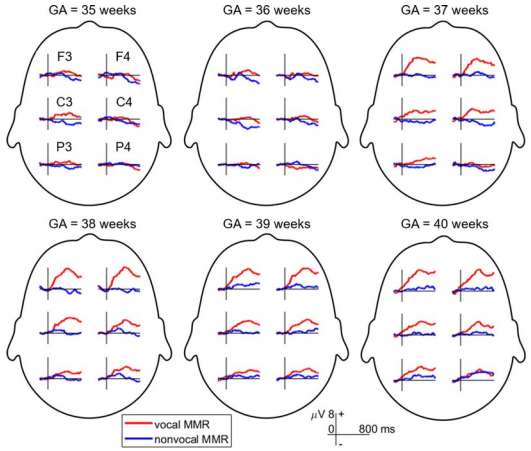
Carlson and Birkett (2022), Fig. 14.7, p. 463.

Newborns discriminate vocal emotion

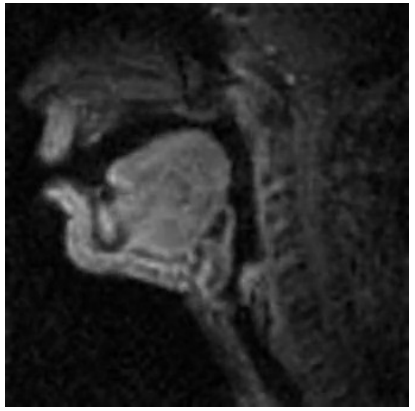
A study recorded electrical brain responses in sleeping newborns presented with syllables spoken in happy or neutral tones. Infants were grouped by gestational age, meaning the developmental time since the beginning of pregnancy. The experiment examined when a neural difference between emotional vocal expressions became detectable, rather than asking newborns to provide an explicit judgment about emotion.

The researchers used an oddball sequence in which one type of stimulus occurred regularly and another appeared less often. Subtracting the response to the regular sound from the response to the different sound yielded a mismatch response. This response measures neural discrimination of the change; it does not establish that the infant can name or consciously explain the speaker's emotion.

In the reported groups, a reliable difference emerged around 37 weeks of gestational age and was not observed in the younger groups under the tested conditions. The finding concerns a developmental change in the measured response during natural sleep. It does not imply that all aspects of social understanding begin at one precise age or develop at the same rate.



Hou et al. (2024), Fig. 1.



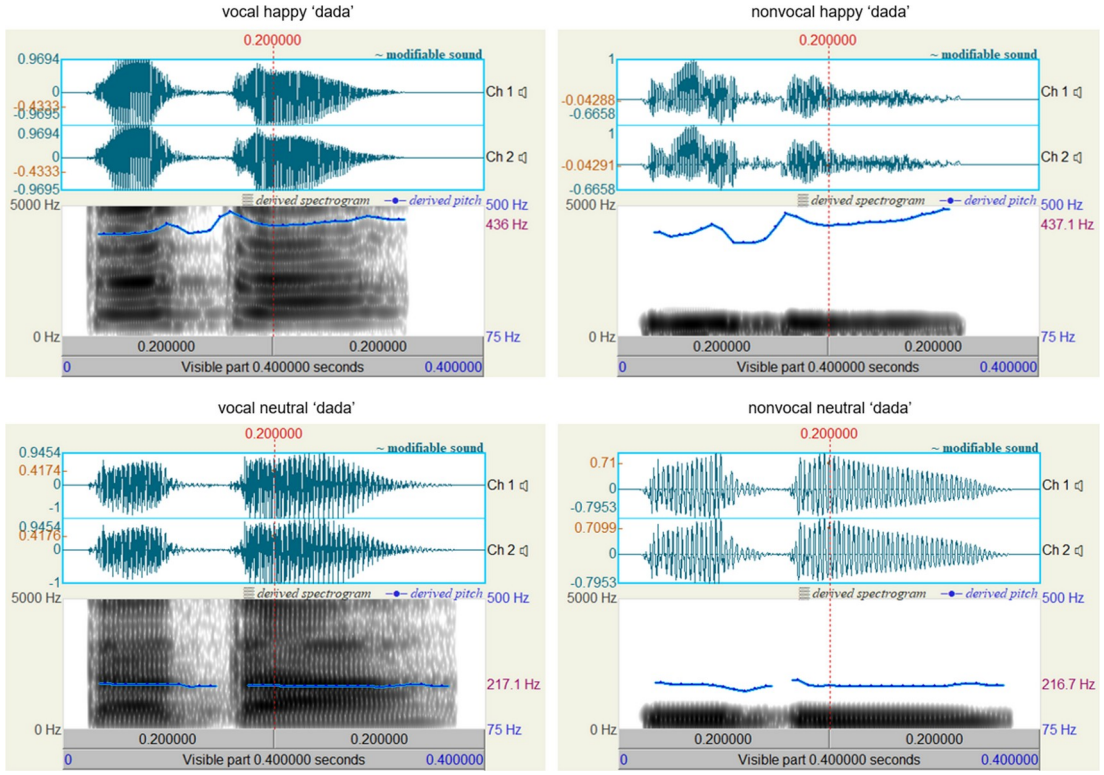
USC SPAN / Esling (2015), rtMRI: hood.

Nonvocal controls isolate sensitivity to voices

The newborn prosody study also used nonvocal control sounds constructed to preserve relevant acoustic properties of the vocal stimuli. This comparison asked whether the developmental response could be explained simply by differences in the physical signal. The original stimulus figure shows the sound waveforms and spectrograms, which display how acoustic energy changes across time and frequency.

The older newborn groups discriminated the happy and neutral vocal expressions without showing the same pattern for the matched nonvocal comparison. The result therefore favored sensitivity to the organization of vocal emotion over a response to every acoustic difference present in the stimuli. Matching acoustics provides a stronger test than comparing unrelated emotional and neutral recordings.

An electrical mismatch response is a physiological measure of discrimination, not a diagnostic test for an individual child's future social development. The experiment identified a response pattern in groups of healthy newborns. Its biological contribution is to specify when and under which stimulus conditions sensitivity to emotional vocal information could be detected in the developing auditory system.



Hou et al. (2024), Fig. 3.

Aphasia affects language components differently

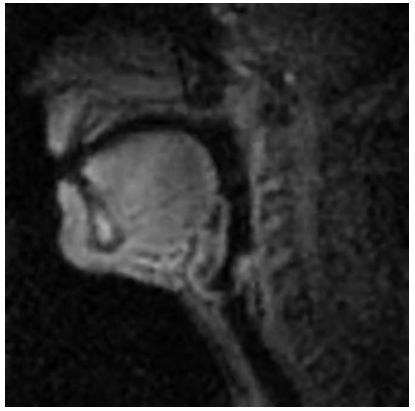
Aphasia is an acquired disturbance of language following brain injury or disease. It can affect naming, sentence formation, comprehension, repetition, reading, and writing in different combinations. The diagnosis concerns language processing rather than speech absence alone. Deafness, paralysis of the vocal tract, and reduced alertness can each prevent a spoken response through different mechanisms.

Clinical assessment compares several tasks because apparently fluent speech can contain severe word selection and comprehension errors. Conversely, slow effortful speech can coexist with substantial understanding. A picture description reveals how a person selects and combines information, but additional naming, repetition, and comprehension tasks are needed to identify which processes remain available.

Classic aphasia categories summarize common patterns rather than perfectly bounded lesions. Damage can include cortex and the white matter pathways connecting it to other regions. Recovery also changes performance over time. Describing the actual abilities of the person provides a more precise account than assuming that a diagnostic label identifies one isolated damaged language center.



Carlson and Birkett (2022), Fig. 14.8, p. 465.



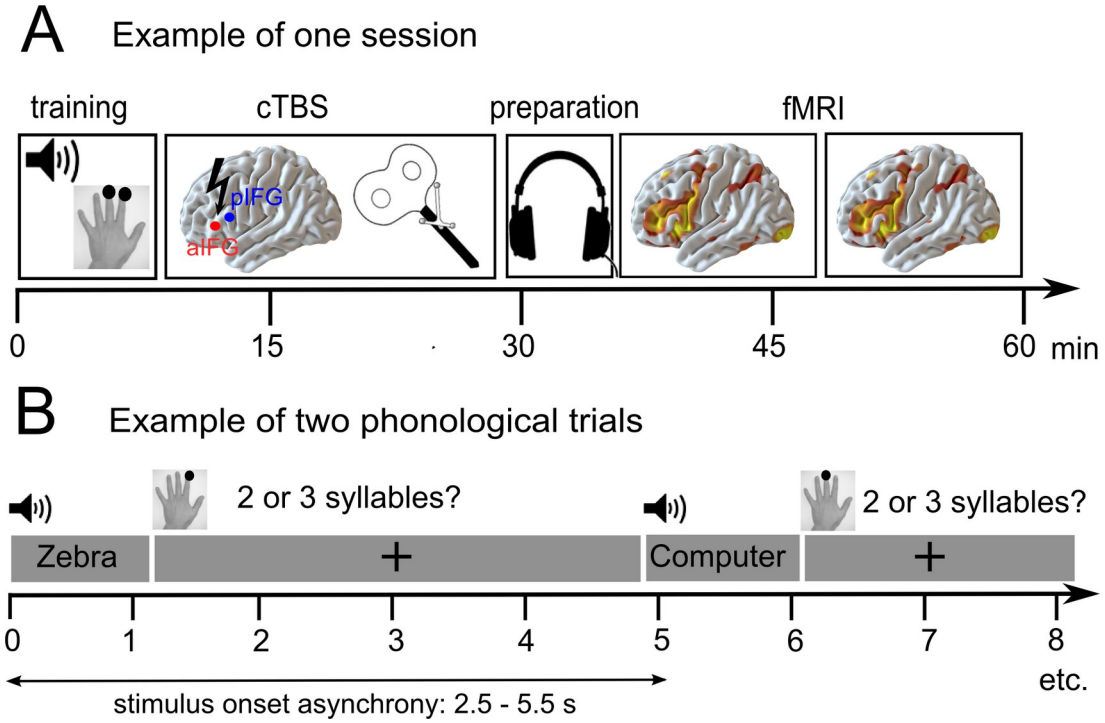
USC SPAN / Esling (2015), rtMRI: head.

Perturbation tests the poststroke language network

A study examined people with left temporoparietal stroke lesions using transcranial magnetic stimulation followed by functional imaging. Magnetic stimulation briefly changes activity in the targeted cortical region without producing a permanent lesion. The researchers stimulated anterior or posterior left inferior frontal cortex and compared the consequences with a sham condition that controlled for the procedure.

Participants made judgments about word sounds and meanings. The investigators asked whether perturbing a surviving frontal language region would disrupt performance and recruit other parts of the network. The design tested the short term response of a brain already altered by stroke, rather than inferring recovery solely from a comparison between patients and healthy participants.

Phonological judgments concern how words sound, while semantic judgments concern their meanings. Testing both made it possible to identify a selective effect instead of treating language as one undivided capacity. The stimulation sites, behavioral tasks, and subsequent imaging together specified which network operation was challenged and how the remaining system responded.



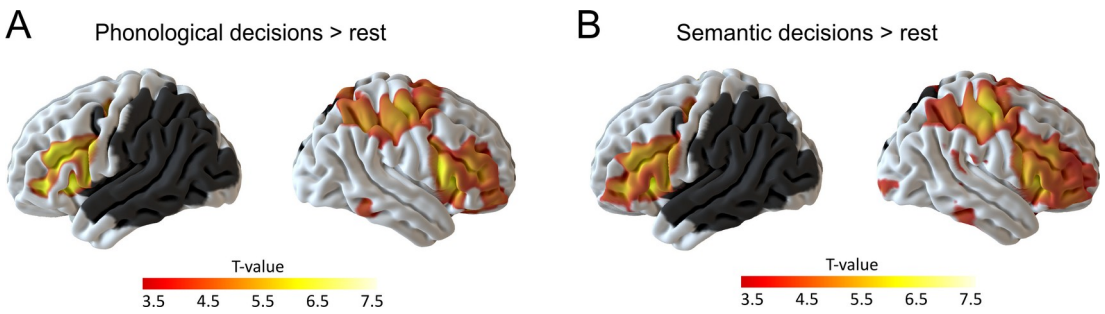
Hartwigsen et al. (2020), Fig. 1.

Right hemisphere recruitment can support performance

In the poststroke perturbation study, stimulation of posterior inferior frontal cortex selectively delayed phonological decisions and reduced activity associated with that task. The delay indicated that the surviving left frontal region still contributed to performance. A lesion elsewhere in the language network therefore had not made this residual frontal tissue irrelevant to processing word sounds.

People who showed greater disruption also showed stronger recruitment of the corresponding region in the right hemisphere. Better integrity of a right hemisphere white matter pathway was associated with less disruption. These functional and structural measurements supported a compensatory contribution under the particular task and stimulation conditions rather than a universal benefit from all right hemisphere activation.

The sham condition identified the residual network before effective perturbation, while the stimulation comparison tested its capacity to adapt. Increased activity after injury can have several causes, so its role must be related to behavior. Here, the specific pairing of response delay, regional recruitment, and tract integrity gave a more informative account than activation alone.



Hartwigsen et al. (2020), Fig. 3.



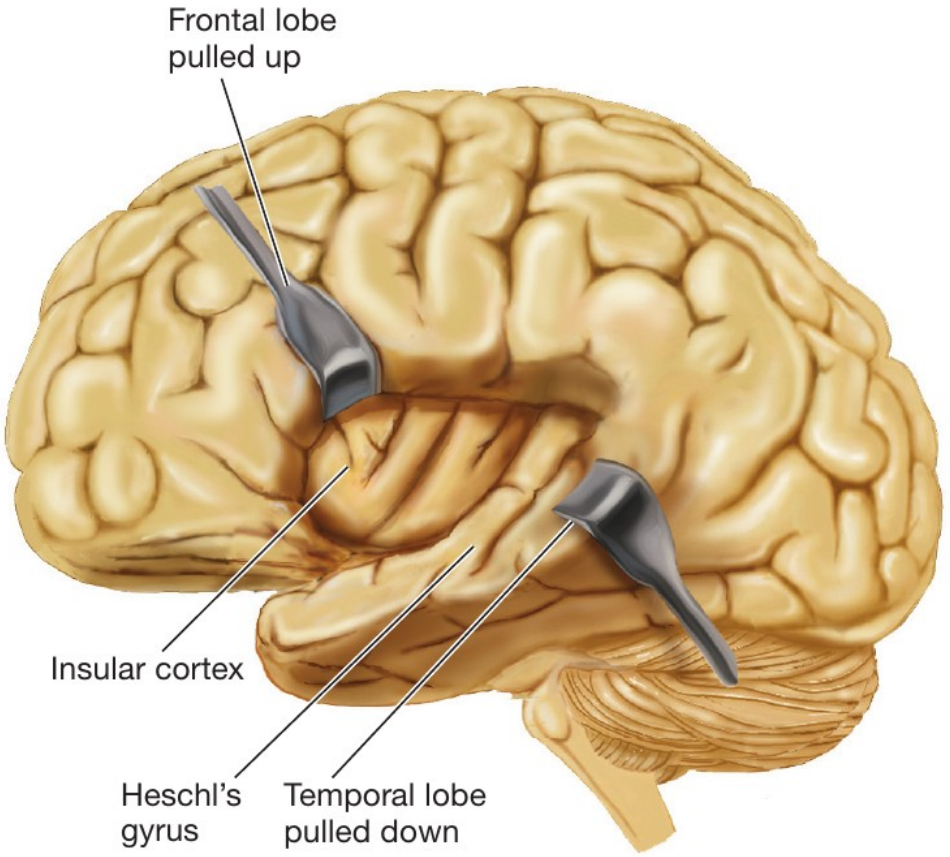
USC SPAN / Esling (2015), rtMRI: had.

Broca type aphasia disrupts sentence production

Broca type aphasia commonly includes slow, effortful output, difficulty finding words, and reduced grammatical structure. A person may produce a short sequence of important content words while omitting small grammatical words or endings. This is called agrammatism. The intended concept can be clearer than the sentence that the person is able to formulate aloud.

Comprehension is often better than spontaneous production, but it is not necessarily normal. Sentences whose meaning depends heavily on grammatical relationships can be difficult, especially when word order must identify who acted on whom. A person may understand the separate nouns and verbs while struggling to assign the roles required by a particular sentence construction.

The syndrome can follow damage extending beyond the small frontal territory historically called Broca’s area. Adjacent premotor tissue, insula, and connecting pathways can contribute different symptoms. Speech planning difficulties may coexist with language formulation problems. Separating grammar, naming, and articulatory sequencing explains why two people with a similar diagnostic label can need different forms of communication support.



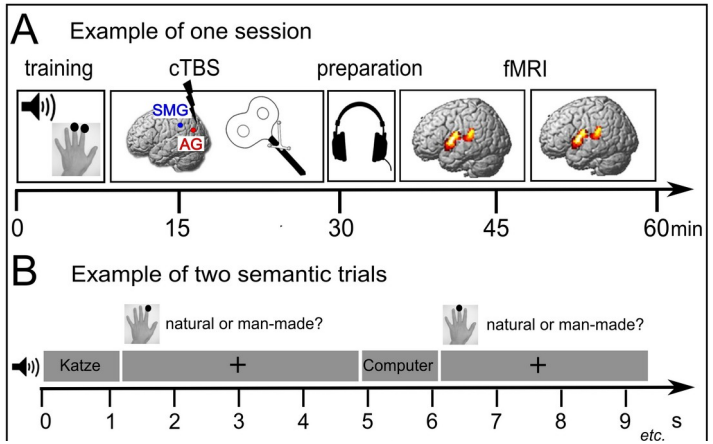
Carlson and Birkett (2022), Fig. 14.9, p. 467.

Different perturbations elicit different adaptation

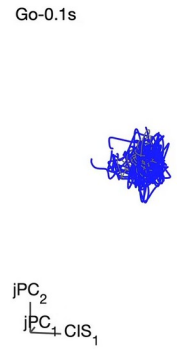
A study of healthy participants used temporary magnetic stimulation to perturb either the angular gyrus or the supramarginal gyrus in the left parietal lobe. These regions participate differently in semantic and phonological processing. Participants then performed meaning and sound judgments during functional imaging, allowing the researchers to compare behavioral disruption with changes across the broader language network.

Perturbing the semantic region reduced activity in a distributed semantic network and increased recruitment of neighboring areas usually associated with phonological processing. Perturbing the phonological region also disrupted behavior and suppressed network activity, but did not produce the same compensatory upregulation. The network response therefore depended on which operation had been challenged.

The experiment measured rapid reorganization after a reversible manipulation rather than long term recovery from a stroke. It provides a way to test how alternative regions become involved when normal processing is disturbed. The results do not require assuming that a neighboring region permanently takes over the damaged function after every form of brain injury.



Hartwigsen et al. (2017), Fig. 1.



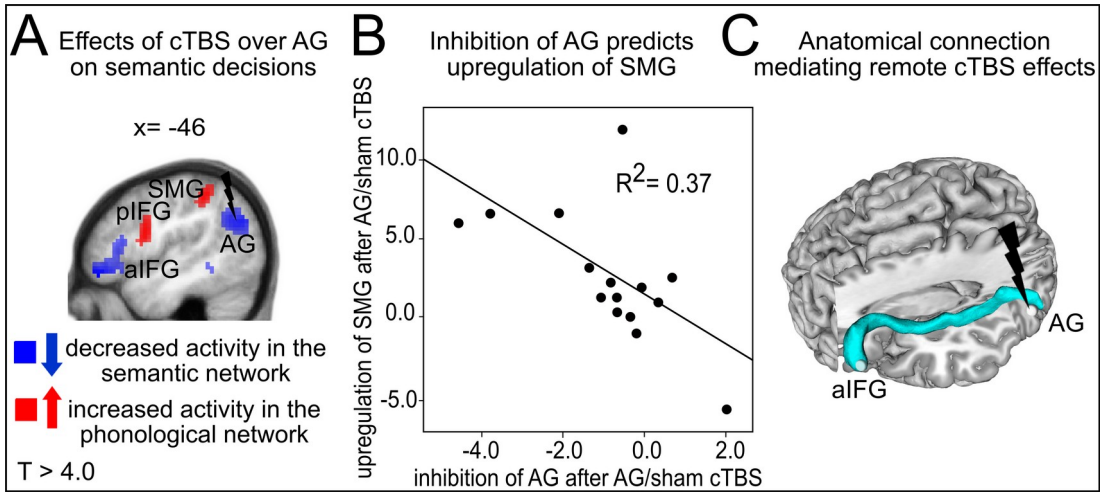
Stavisky et al. (2019), Video 2.

Network influences predict semantic slowing

After perturbation of the angular gyrus, the language reorganization study found a change in its effective influence on another semantic region. Effective connectivity is an analysis of directed interactions under a specified model. The disrupted region exerted a stronger inhibitory influence, and the estimated influence predicted how much the participant's semantic decisions slowed.

People with greater behavioral disruption also showed more semantic recruitment of neighboring phonological territory. The combination identified both a source of interference and a possible route for adaptation within the network. An increased response in a nearby region was therefore interpreted alongside the person's performance and the modeled interactions among the regions engaged by the task.

The analysis concerns changes in functional interactions during the experiment, not a direct measurement of inhibitory synapses between every pair of cortical areas. Magnetic stimulation, imaging, and computational modeling contributed different evidence. Together they specified which manipulation changed processing and how those changes related to the timing of a meaningful language response.



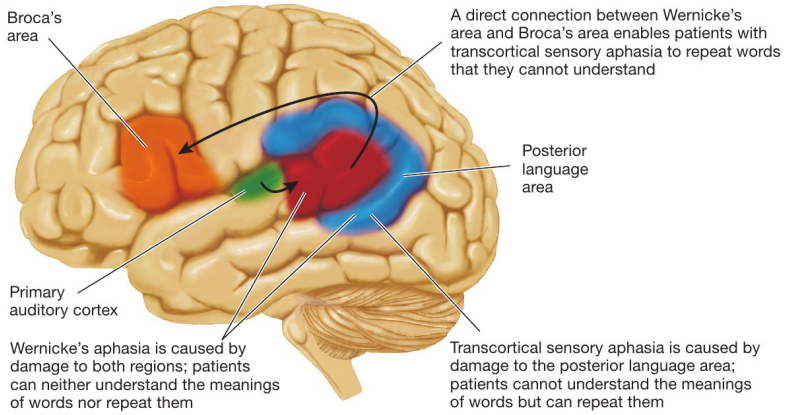
Hartwigsen et al. (2017), Fig. 3.

Fluent speech can coexist with impaired meaning

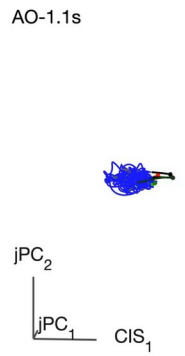
Wernicke type aphasia often includes fluent speech with incorrect words, sound substitutions, or invented forms, together with impaired comprehension. The person may produce a normal sounding rhythm and sentence length while conveying little of the intended message. Preserved fluency therefore does not establish that word meanings are being retrieved and monitored normally.

Posterior temporal injury can impair analysis of spoken word forms, while injury to other temporal and association regions can impair access to conceptual meaning. Some people can repeat a word that they cannot understand, demonstrating a relatively preserved route from heard speech to spoken output. Others fail both repetition and comprehension because additional processing stages are affected.

Semantic knowledge can also deteriorate gradually in diseases affecting anterior temporal cortex. This differs from an abrupt posterior temporal stroke, although both can impair word understanding. Comparing acquired syndromes helps identify the damaged process: recognizing the incoming word, accessing its meaning, selecting an appropriate response, or monitoring the resulting speech.



Carlson and Birkett (2022), Fig. 14.14, p. 471.



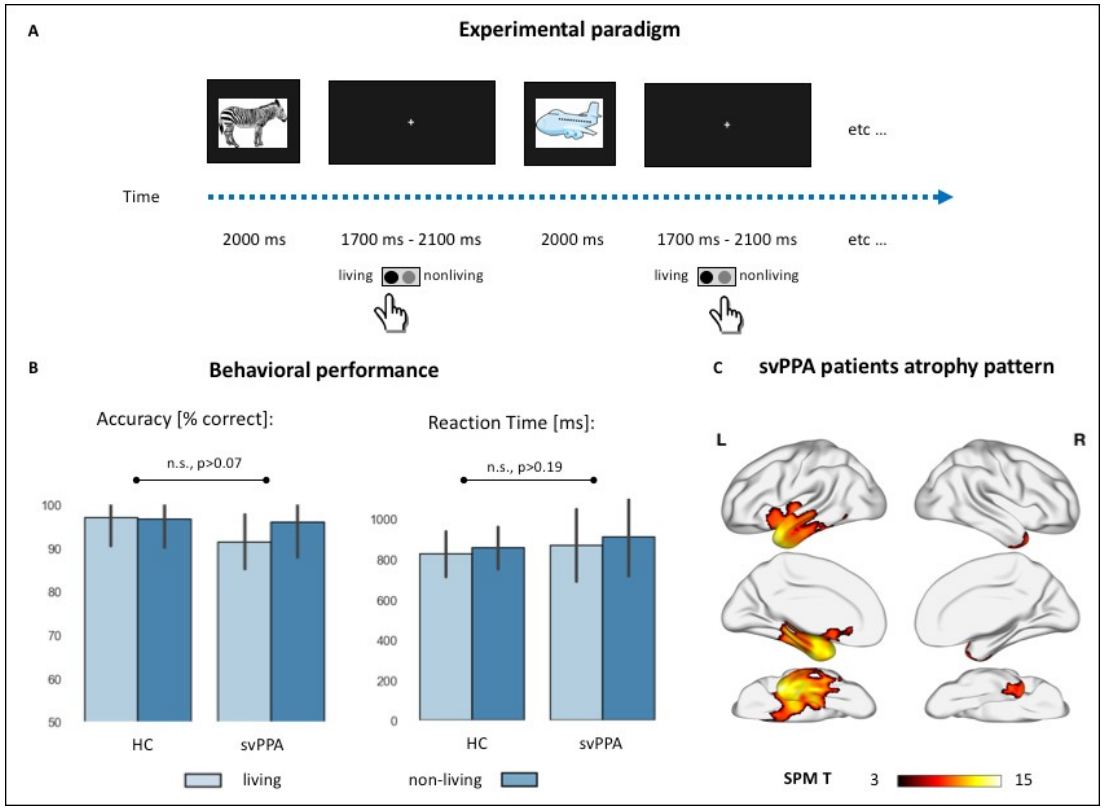
Stavisky et al. (2019), Video 3.

Temporal degeneration disrupts semantic knowledge

A study examined semantic variant primary progressive aphasia, in which degeneration affects anterior temporal cortex. Participants and healthy volunteers categorized concepts during magnetoencephalography. The researchers compared brain responses while the groups performed the same task, asking how the remaining network handled meaning after degeneration compromised an important component of the conceptual system.

The anterior temporal lobe contributes to organizing conceptual knowledge across experiences and sensory features. Damage in this syndrome can cause progressive loss of word and object meanings. The experiment therefore concerned a different biological disturbance from impaired muscle control or difficulty hearing speech, even though each can make everyday communication more challenging.

Performance on the particular categorization task was similar across the reported groups, allowing neural differences to be examined when an overt answer remained available. A correct response does not imply that the brain used its usual processing route. Additional perceptual information can help a person solve a task even when the conceptual system is less efficient or less complete.



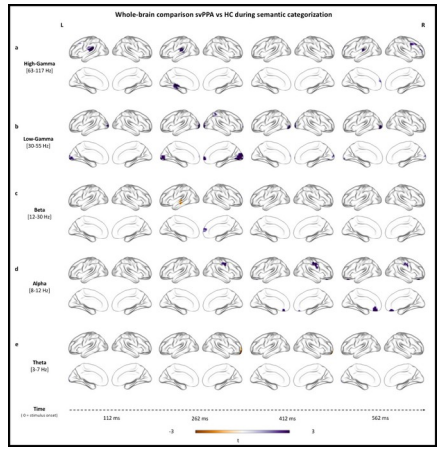
Borghesani et al. (2021), Fig. 1.

Semantic tasks recruit altered perceptual processing

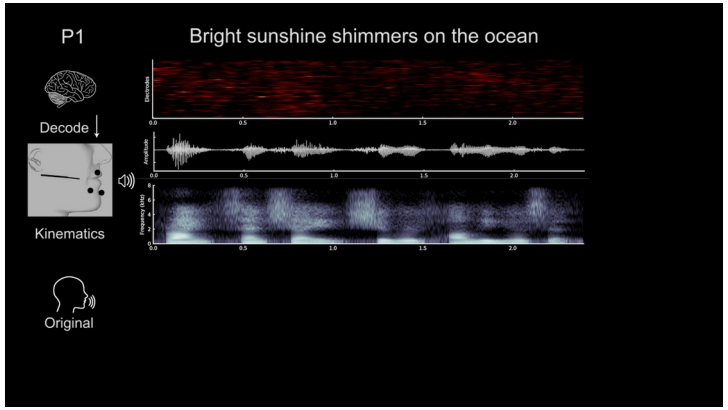
The semantic variant aphasia study found greater engagement of bilateral occipital cortex and superior temporal regions during the categorization task, alongside less consistent frontal recruitment. These areas contribute to perceptual analysis and language related processing. Their altered activity described how the network operated after anterior temporal degeneration rather than locating all semantic knowledge in one newly active region.

The recorded oscillatory changes varied across frequency bands and task periods. Magnetoencephalography detects the timing of coordinated neural activity, so the analysis could distinguish changes as conceptual information was being processed. A whole trial average would conceal some of those differences and could make two networks appear similar despite different sequences of regional engagement.

The authors interpreted the increased perceptual contribution as a way of compensating for degraded semantic representations. The task's available features could support an answer even when access to broader conceptual knowledge was altered. This helps explain why preserved performance on one categorization test can coexist with substantial difficulty understanding and using words in everyday communication.



Borghesani et al. (2021), Fig. 3.



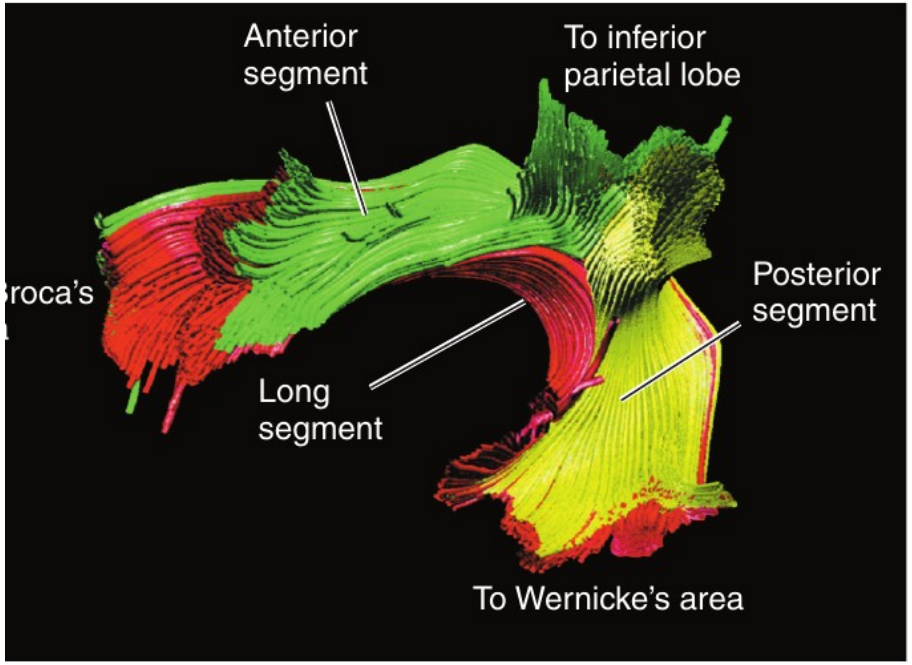
Anumanchipalli et al. (2019), Video 1.

Conduction aphasia especially impairs repetition

Conduction aphasia commonly includes difficulty repeating speech that is disproportionate to some other language abilities. A person may understand a message and speak relatively fluently while making sound errors when asked to repeat it. The problem is especially apparent for unfamiliar sequences that cannot be reconstructed easily from meaning or long established vocabulary.

The arcuate fasciculus and neighboring dorsal pathways connect temporal and parietal language regions with frontal speech systems. Damage can interfere with translating an incoming sound sequence into an organized articulatory response. Contemporary anatomy includes multiple direct and indirect routes, so conduction aphasia should not be reduced to the severing of one universally defined cable.

Meaning provides an alternative source of information for familiar utterances. Someone may paraphrase an instruction correctly while failing to reproduce its exact sounds. Comparing comprehension, spontaneous speech, and repetition therefore distinguishes understanding a message from preserving its phonological sequence long enough to produce a matching response through the speech motor system.



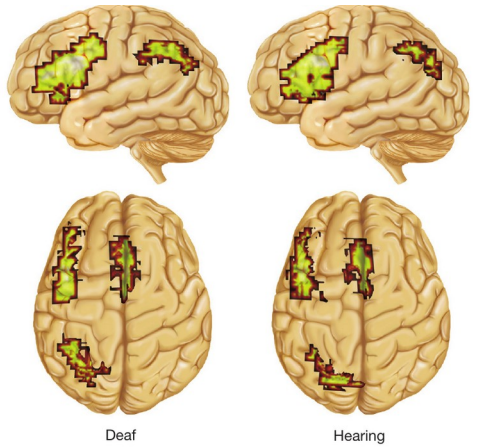
Carlson and Birkett (2022), Fig. 14.18, p. 474.

Signed languages use linguistic networks

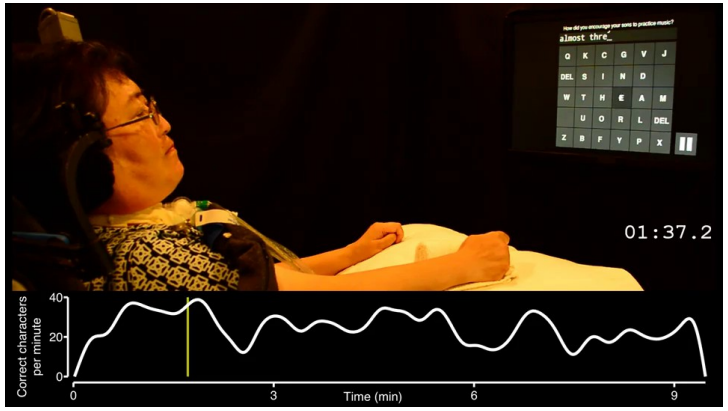
Signed languages are natural languages with vocabulary and grammatical structure, expressed through coordinated hand, arm, facial, and bodily movements. They are not simply pantomime or spoken words converted one at a time into gestures. Their visual input and manual output recruit sensory and motor systems while linguistic operations also engage networks used in other forms of language.

Damage to the dominant hemisphere can produce aphasia in a person who uses a signed language, affecting sign production or comprehension beyond any difficulty moving the hands. A person may retain nonlinguistic gestures while struggling to organize a grammatical signed message. This dissociation separates a language impairment from a general inability to perceive or perform movements.

Early access to an accessible natural language is also distinct from hearing itself. A deaf child exposed to fluent signing can acquire language through vision. Comparing deaf adults with different histories of early sign language exposure allows researchers to investigate how linguistic experience contributes to conceptual processing while holding important aspects of auditory experience more similar.



Carlson and Birkett (2022), Fig. 14.20, p. 476.



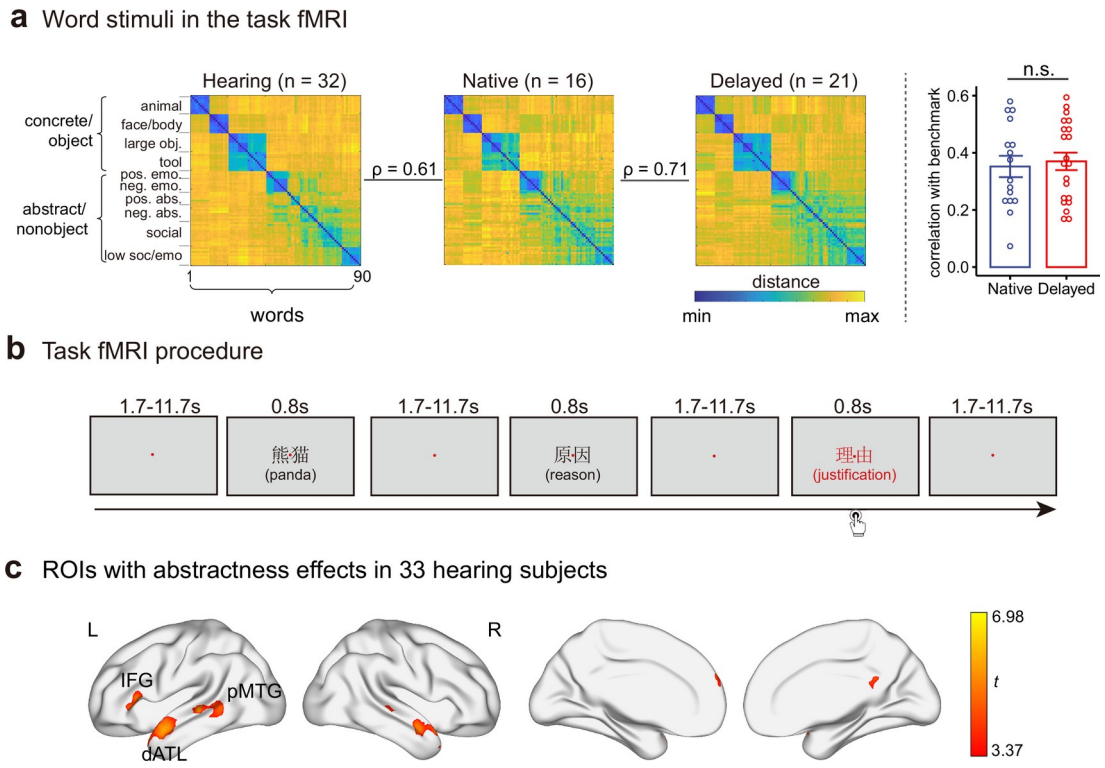
Pandarinath et al. (2017), media1.

Early language access shapes semantic organization

A functional imaging study compared deaf adults who acquired sign language from birth with deaf adults who had reduced early access to any natural language. Both groups performed semantic judgments on written words familiar to them. This comparison examined the consequences of early linguistic experience while reducing the confound between language exposure and the presence or absence of hearing.

The word set included concrete object concepts and abstract concepts not tied to a single visible object. Researchers compared patterns of neural activity with measures of semantic relationships among those words. The analysis asked whether concepts with similar meanings also produced similar activity patterns, rather than only whether reading caused a general increase in blood oxygen related signal.

The left dorsal anterior temporal lobe showed reduced semantic sensitivity in the delayed language group. The result concerns access to early language, not a claim that deafness prevents conceptual knowledge. The native signing comparison group demonstrated that a fully accessible visual language can support semantic organization without the auditory experiences available to hearing children.



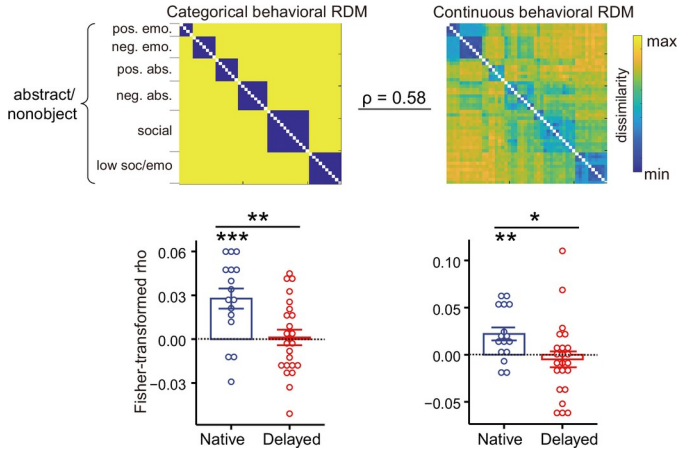
Wang et al. (2023), Fig. 1.

Abstract concepts reflect early language experience

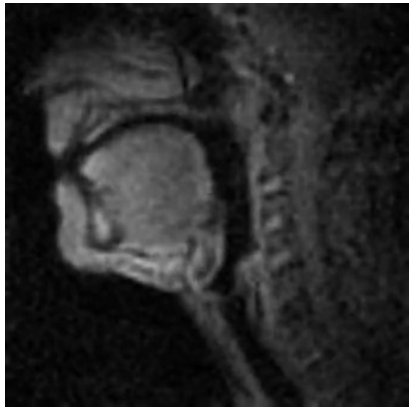
The early language exposure study examined abstract concepts separately from concrete object words. Abstract concepts often depend strongly on relations conveyed through language rather than on one stable visual appearance. The researchers compared how well neural patterns in dorsal anterior temporal cortex reflected semantic relationships among those concepts in native signers and delayed signers.

Delayed signers showed weaker correspondence between neural patterns and semantic structure, even for words familiar to both groups. The experiment also found differences in sensitivity to abstractness. Familiarity with an individual written label therefore did not remove every consequence of an earlier period with reduced access to a natural language and its conceptual relationships.

The comparison was based on language histories rather than an experiment that withheld language from children. Its interpretation depends on the carefully matched sensory background and measured semantic task. The practical developmental distinction remains clear: early accessible signing provides linguistic experience, while absence of auditory input and absence of language are different conditions that should never be treated as equivalent.



Wang et al. (2023), Fig. 3.



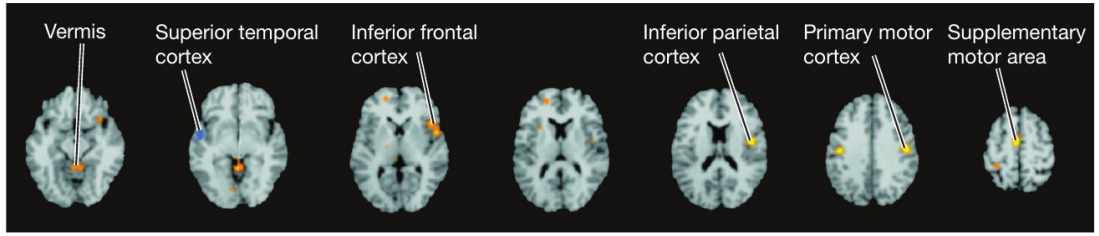
USC SPAN / Esling (2015), rtMRI: hid.

Stuttering interrupts speech timing and continuity

Developmental stuttering involves repetitions, prolongations, or blocks that interrupt the flow of speech. A person can know the intended word and its meaning while having difficulty initiating or continuing its spoken production. Stuttering therefore differs from an inability to understand language, and its occurrence does not indicate reduced intelligence or a lack of effort.

Speech fluency depends on coordinating initiation, sequencing, and sensory feedback across motor, auditory, and subcortical circuits. Some people show changes in fluency when speaking with another person, singing, or receiving altered auditory feedback. These conditions change the timing information available to the system and reveal that the same person’s fluency can vary substantially across speaking situations.

Imaging findings identify differences across groups, but no single activation pattern explains every person’s experience. Developmental history and inherited influences also contribute. Treatment aims to support effective communication and reduce the burden of speaking; the biological study of fluency should distinguish the mechanisms of speech interruptions from assumptions about the speaker’s confidence or understanding.



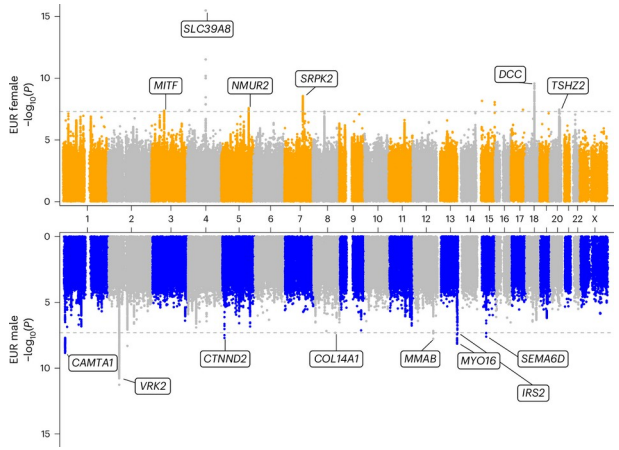
Carlson and Birkett (2022), Fig. 14.21, p. 477.

Many genetic regions contribute to stuttering risk

A large genome wide association study examined genetic variation associated with self reported stuttering. An association study compares the frequency of DNA variants across groups to identify regions that track the trait. The investigators analyzed groups defined by sex and ancestry and then combined evidence across analyses, identifying many associated genomic locations rather than one universal stuttering gene.

The study also compared genetic patterns with those of other traits, including aspects of language, rhythm, and neurodevelopment. Shared genetic variation can indicate overlapping biological influences without making the conditions identical. A variant associated with risk does not specify which individual will stutter, how severe the interruptions will be, or which treatment will help that person.

The original genomic plot shows the distribution of association evidence across chromosomes. Each point represents a tested genetic variant, not a neuron or an anatomical speech center. The results describe a distributed genetic architecture in which many effects contribute to probability, consistent with the diversity of developmental histories and clinical presentations among people who stutter.



Polikowsky et al. (2025), Fig. 1.



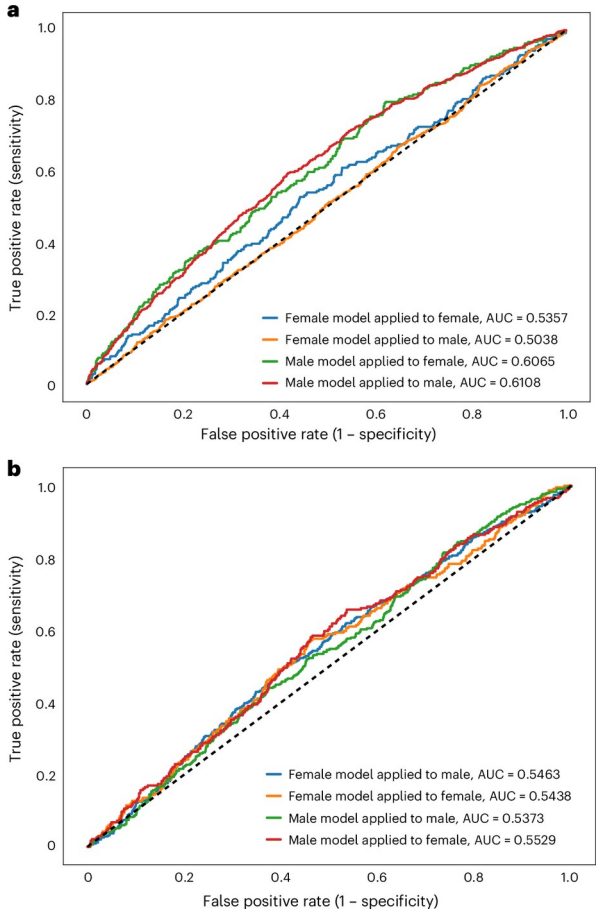
USC SPAN / Esling (2015), rtMRI: hayed.

Genetic scores estimate stuttering probability

The stuttering genetics study evaluated polygenic risk scores in independent datasets. A polygenic score combines many DNA variants, weighted by their associations in a discovery sample. Testing it in new participants asks whether the combined information generalizes beyond the group used to estimate those weights rather than merely reproducing the original statistical comparison.

The scores carried information about stuttering risk in the validation datasets, supporting a contribution from many common variants. However, their biological meaning remains probabilistic. A score summarizes inherited associations in the sampled populations; it does not measure the moment to moment neural events that produce a block, repetition, or prolongation during a particular utterance.

The analysis also depended on how stuttering was identified and on the ancestry represented in the source data. Those features determine what a score can predict and in whom it has been evaluated. Genetic evidence therefore complements behavioral and neural assessment while leaving the person's actual communication abilities and needs to be measured directly.



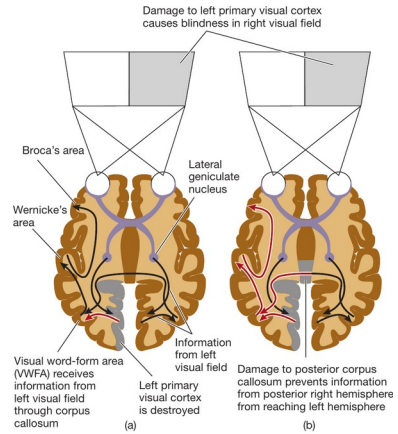
Polikowsky et al. (2025), Fig. 4.

Pure alexia impairs visual access to words

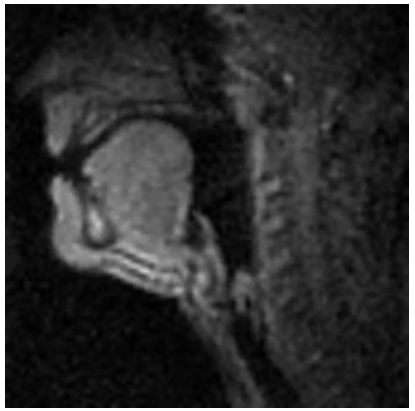
Alexia is an acquired reading impairment, and pure alexia refers to a pattern in which reading is severely affected despite relatively preserved writing. A person may write a message but then be unable to read it back normally. This dissociation shows that producing an orthographic sequence and recognizing the same sequence through vision require different transformations.

One classic lesion pattern damages left occipital cortex and posterior callosal fibers. Visual information from the remaining visual cortex then has difficulty reaching left hemisphere language related systems. Other lesions can directly affect ventral occipitotemporal regions important for efficient recognition of written words. The anatomical account must therefore match the particular person's visual and language findings.

Some affected readers identify letters slowly and assemble a word one letter at a time. The resulting increase in reading time with word length helps characterize the impaired operation. Preserved spoken comprehension and writing do not make the reading difficulty less real; they reveal which components remain available when visual word recognition has been disrupted.



Carlson and Birkett (2022), Fig. 14.24, p. 480.



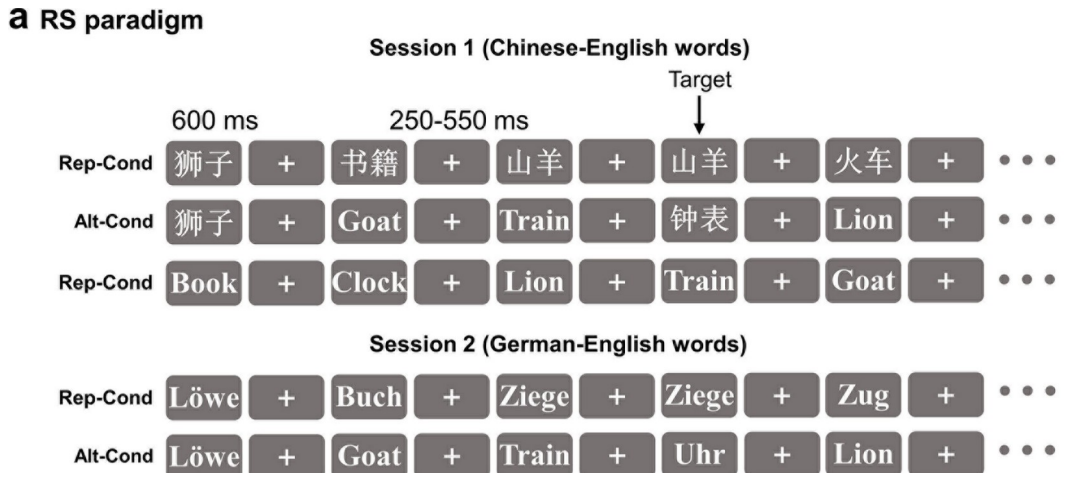
USC SPAN / Esling (2015), rtMRI: bead.

Neural responses distinguish writing systems

A study tested how the brain categorizes visual words from different languages and writing systems. Participants viewed sequences that either repeated one language or alternated between languages. Electrical and magnetic recordings measured responses to those sequences. The comparison asked whether language category changed neural processing even when the immediate task did not require translating each word into speech.

Repetition suppression is a reduction in response when related stimuli are repeatedly presented. Here, differences between repeated and alternating sequences provided evidence that neural populations distinguished categories of written language. The researchers combined average response measurements with analyses of the similarity between activity patterns to examine how those distinctions were represented across the brain.

The experiment included alphabetic and nonalphabetic writing systems and participants with different language backgrounds. Comparing these conditions separated some effects of visual language category from particular word meanings or pronunciations. Written symbols can therefore carry category information in addition to the semantic and phonological information used when a reader interprets a specific message.



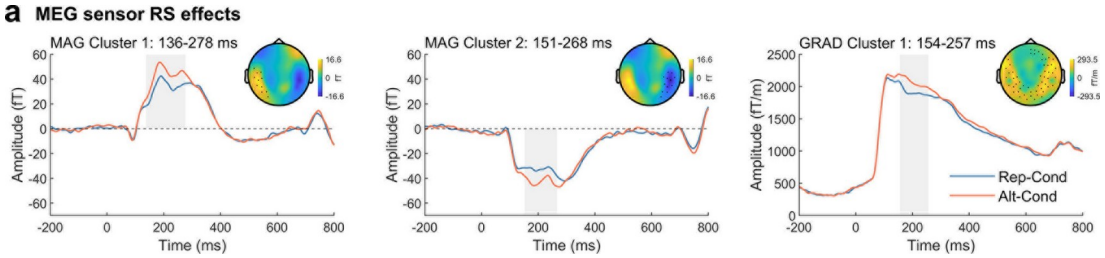
Zheng et al. (2026), Fig. 1.

Visual language categories engage a broad network

The visual word categorization study found early category related responses in a network including ventral occipitotemporal, anterior temporal, insular, and orbitofrontal regions. The reported effects appeared from roughly 150 milliseconds after the stimulus. This timing indicates that category information can emerge rapidly, before a deliberate spoken explanation of the word would be completed.

The category effects were observed across participants with different linguistic backgrounds and were separable from the tested meanings and pronunciations. The analysis also identified interactions within hemispheres. Thus, even a relatively simple distinction between writing systems involved distributed processing rather than a single visual switch that labels every unfamiliar script in exactly the same way.

These measurements concern neural discrimination of category, not successful comprehension of every displayed word. A person can recognize that a string belongs to a writing system without knowing its meaning. That distinction parallels the clinical separation between seeing recognizable letter patterns, accessing their pronunciation, and recovering the conceptual content of the written message.



Zheng et al. (2026), Fig. 3.



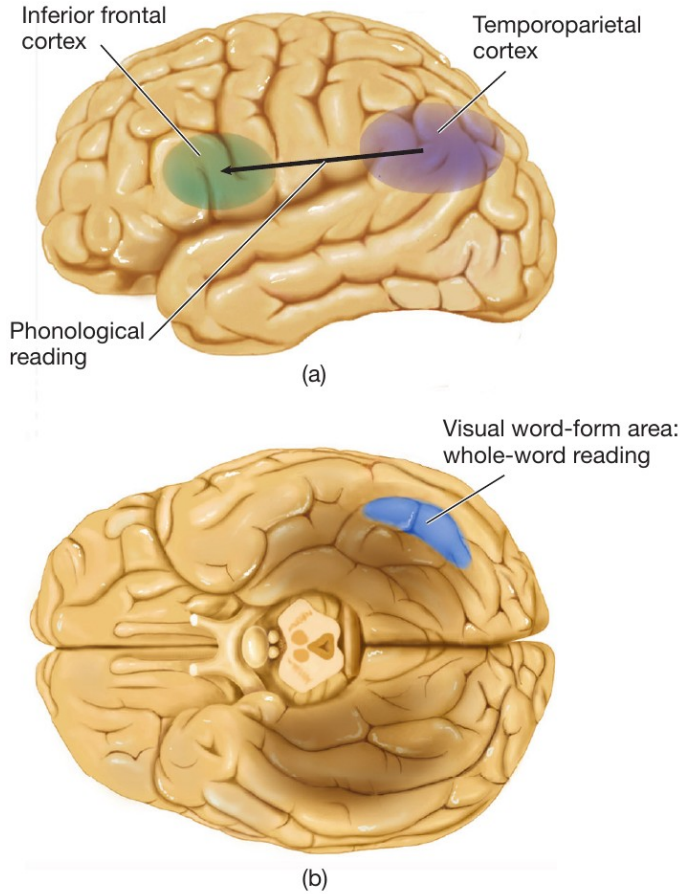
USC SPAN / Esling (2015), rtMRI: bid.

Reading combines visual sound and meaning routes

Reading begins with visual analysis of marks and proceeds through learned representations of letters, character combinations, and words. Ventral occipitotemporal cortex becomes specialized through experience for useful visual regularities in writing. This region interacts with temporal, parietal, and frontal systems that contribute to pronunciation and meaning rather than completing the entire reading process in isolation.

One route uses learned correspondences between written units and speech sounds. It is especially useful for unfamiliar words and pronounceable nonwords. Another route uses knowledge of familiar written words, including irregular spellings whose pronunciation cannot be recovered reliably from simple sound rules. Skilled reading uses interacting sources of information rather than choosing one route for every situation.

The meaning of a written word can also be accessed without saying it aloud. A reader may understand a sentence silently while motor systems remain relatively quiet. Comparing familiar words, irregular words, and nonwords reveals which operations support performance and which are impaired when acquired injury or developmental difficulty disrupts the reading network.



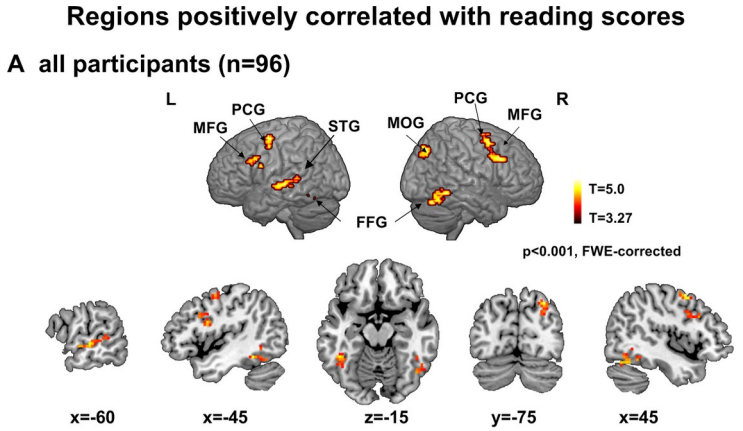
Carlson and Birkett (2022), Fig. 14.30, p. 484.

French and Chinese readers share neural resources

A cross cultural functional imaging study compared children reading French and Chinese, two writing systems with very different relationships between written forms and pronunciation. Children varied in reading skill within each language. The experiment examined responses to words alongside faces and houses, allowing reading related activation to be distinguished from a general response to visual stimuli.

Both language groups recruited a broadly shared network including left fusiform, temporal, precentral, and middle frontal regions. The fusiform portion lies on the ventral surface of the temporal lobe and contributes to visual word processing. Better reading skill was associated with stronger responses in parts of this network across the sampled children.

The shared organization did not erase differences between writing systems. Instead, the comparison asked whether language and reading ability modulated overlapping regions. Studying both factors in the same design provides a stronger test than assigning one cortical area to alphabetic reading and a completely separate area to reading a visually different script.



Feng et al. (2020), Fig. 2.



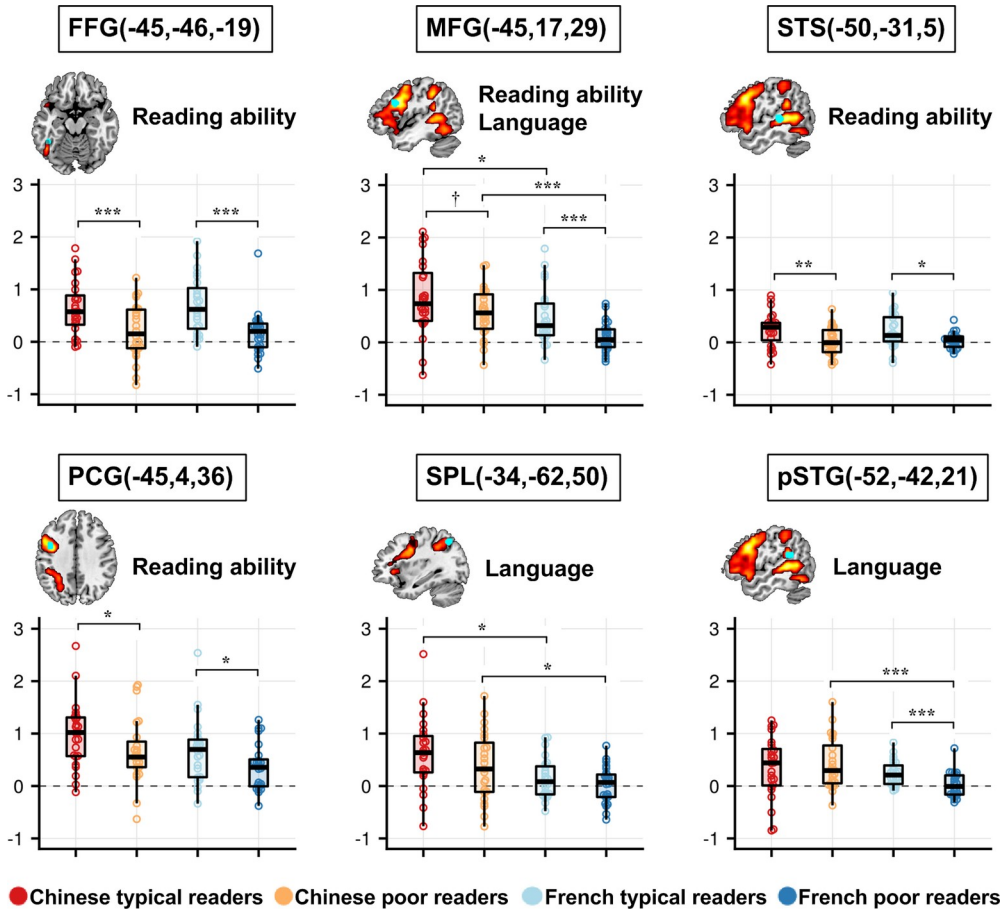
USC SPAN / Esling (2015), rtMRI: bayed.

Reading difficulty affects overlapping regions

In the French and Chinese study, reading ability influenced responses in several regions similarly across the two languages. Some regions also differed according to language, but the effects did not form two wholly separate reading disability networks. A middle frontal region previously described as especially important in Chinese also showed reading ability effects in the broader comparison.

The investigators examined predefined regions as well as whole brain patterns. In the sampled children, language and reading skill contributed distinguishable changes to activity. This means that a shared neural organization can still adapt to the visual, phonological, and educational demands of a particular writing system rather than being either completely universal or completely language specific.

The study measured children's current reading and brain responses, so it did not follow every developmental step that produced those differences. Its concrete finding is cross cultural overlap in the neural correlates of reading skill and difficulty. The result argues against treating struggling readers of different scripts as having entirely unrelated biological problems.



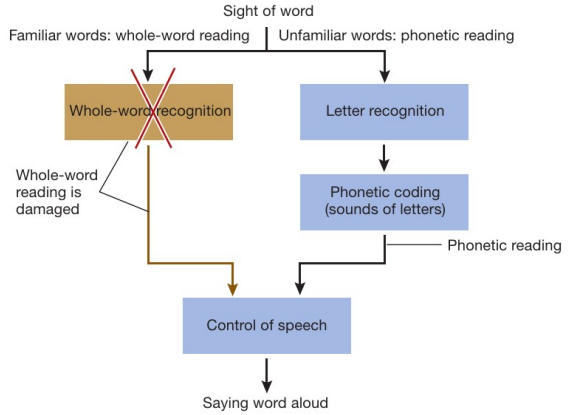
Feng et al. (2020), Fig. 4.

Acquired dyslexias separate reading routes

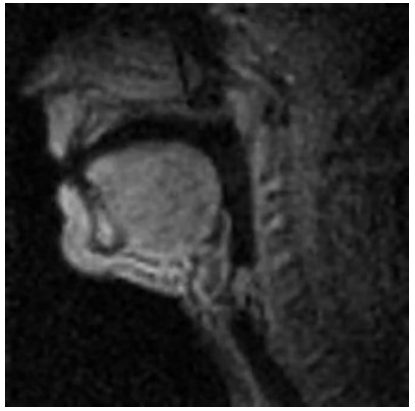
Surface dyslexia particularly affects reading irregular words, which require knowledge of their established pronunciations. A person may pronounce an irregular word as if it followed the most common spelling rules. This pattern indicates that converting letters to sounds can remain useful when access to familiar whole word information is impaired by an acquired lesion or disease.

Phonological dyslexia produces the complementary difficulty: familiar words can be read more successfully than unfamiliar pronounceable strings. The person cannot rely effectively on the sound conversion operations needed for a new sequence. A nonword is informative because it lacks a stored word entry that could otherwise conceal the impaired conversion process during ordinary reading.

Direct reading without comprehension provides another dissociation. A person can produce the pronunciation while failing to retrieve its meaning, showing that an output route remains available without successful semantic access. These syndromes are patterns of acquired impairment; they should not be assumed to describe every child who has developmental difficulty learning to read.



Carlson and Birkett (2022), Fig. 14.27, p. 482.



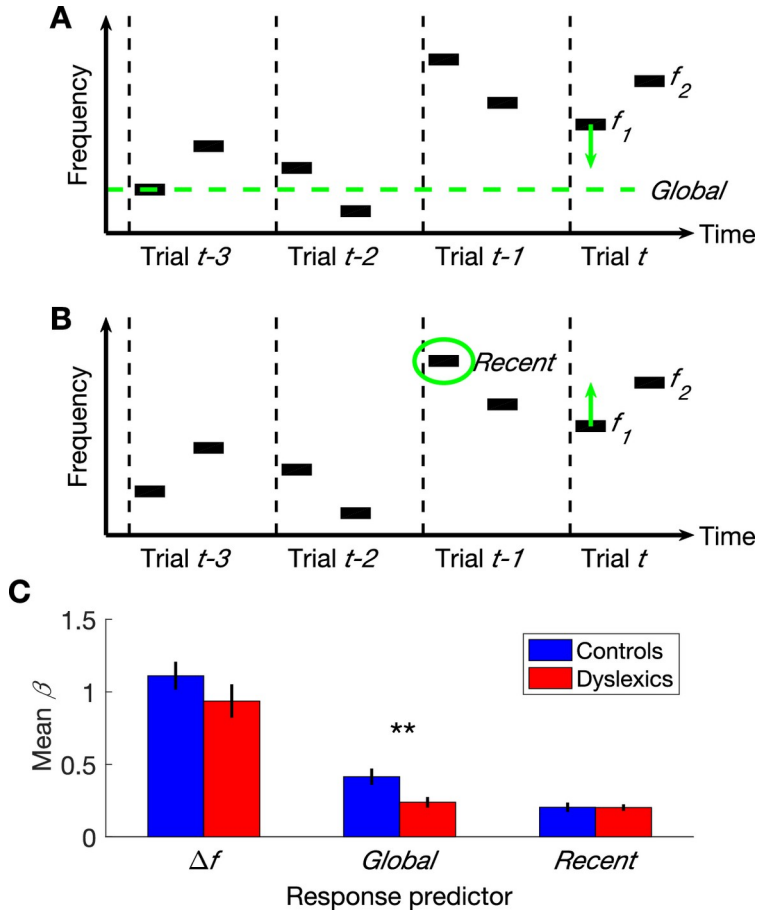
USC SPAN / Esling (2015), rtMRI: bed.

Recent sound history influences perception

A study of adults with dyslexia examined how recent sounds influenced judgments of tone frequency. Participants compared two tones on each trial. Although the immediate task concerned that pair, previous trials also biased the judgment. Such effects reveal implicit use of sensory history, meaning that recent information changes performance without requiring deliberate recollection of the earlier sounds.

The researchers varied the interval between trials and measured both behavioral effects and electrical brain responses. They hypothesized that the influence of recent information decayed more rapidly in dyslexia. A longer delay between trials should then reduce the benefit or bias from previous stimuli sooner in the dyslexic group than in the comparison group.

The task separated the physical difference between the current tones from the context established by earlier tones. Poorer performance need not arise solely from reduced sensitivity to the present sound. It can also arise when the system integrates recent experience over a shorter period and therefore uses less stable information about recurring sensory regularities.



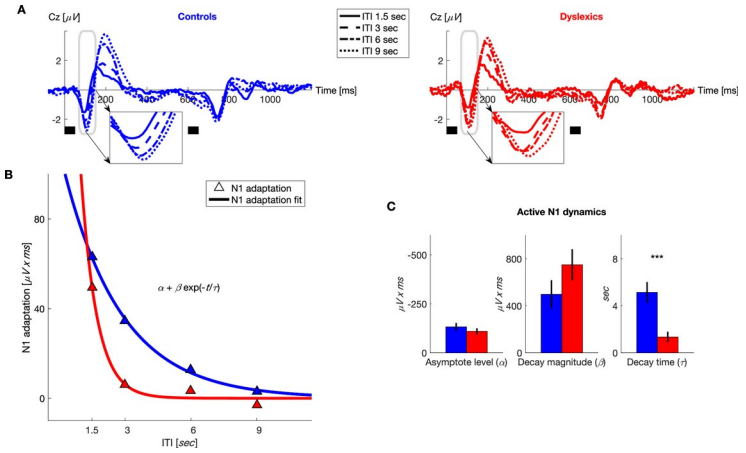
Jaffe-Dax et al. (2017), Fig. 1.

Adaptation decays faster in sampled dyslexic adults

The tone study found a faster decay of adaptation in the dyslexic group's electrical responses. Adaptation is a reduced response after recent stimulation. As the delay grew, the response returned toward its less adapted magnitude sooner in the dyslexic participants, matching a shorter persistence of the effect of earlier stimuli under the tested conditions.

Behavioral influences of recent tones also decayed faster. A related reading experiment found that the benefit from previously reading the same nonword declined more rapidly as the interval increased. These results connected a specific temporal property of sensory history to a task relevant to reading, rather than assuming that tone discrimination alone captures every aspect of dyslexia.

The proposed mechanism concerns a shorter integration window for recent regularities. Less persistent adaptation could reduce the usefulness of repetition when building reliable expectations about sounds and written sequences. This account explains a particular set of findings in the sampled groups; developmental dyslexia remains heterogeneous and cannot be reduced to one measured response in every person.



Jaffe-Dax et al. (2017), Fig. 3.



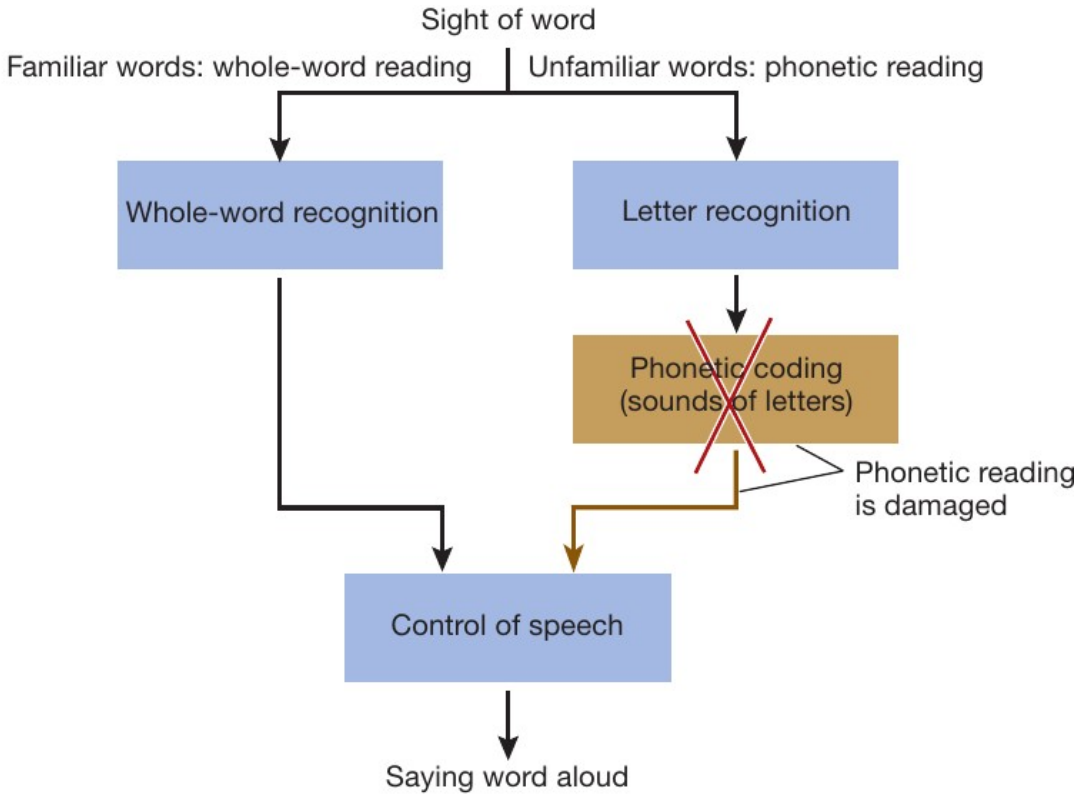
USC SPAN / Esling (2015), rtMRI: bad.

Dyslexia affects the acquisition of fluent reading

Developmental dyslexia is a persistent difficulty with accurate or fluent word reading and spelling that emerges during learning. It is not defined by low intelligence or inadequate effort. Difficulties with phonological processing are common, but readers differ in their profiles, and instruction, language structure, and the opportunities available for practice influence how those difficulties appear.

Learning to read requires connecting visual patterns with an already developing spoken or signed language system. Children gradually improve the speed and reliability of recognizing written units and retrieving their sounds and meanings. A difficulty in one component can slow this learning, while strengths in vocabulary or reasoning may help the child understand material presented through another route.

Brain studies compare groups to investigate mechanisms, but a regional difference on an image is not an individual reading diagnosis. Direct assessment identifies which skills need support. Explicit instruction in the relationships among written forms and language units, together with appropriate access to meaningful material, addresses the operations that a struggling reader must acquire and use.



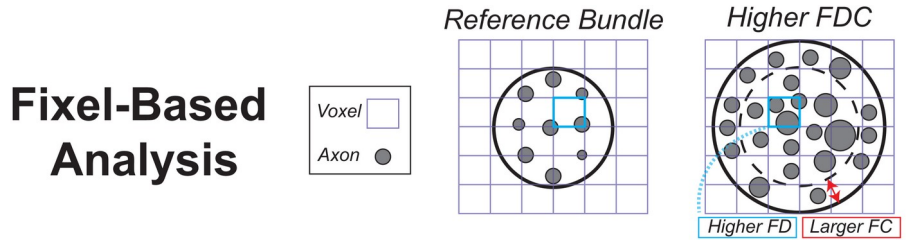
Carlson and Birkett (2022), Fig. 14.28, p. 483.

Diffusion imaging estimates reading related fibers

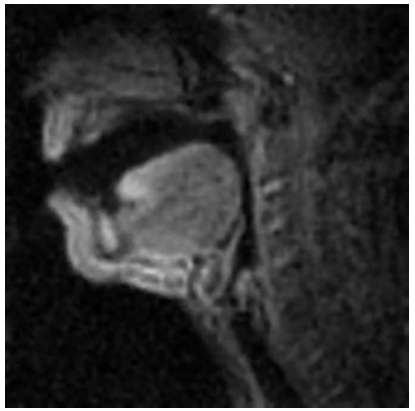
A study of children and adolescents examined whether properties of white matter fibers related to single word reading skill. Diffusion imaging measures the movement of water within tissue. Because water movement is influenced by cellular structure, the method can estimate aspects of fiber organization, but it does not directly photograph each axon or measure individual synaptic strength.

The researchers used a fiber specific analysis that can distinguish multiple fiber directions within one image location. This addresses a difficulty of simpler methods: two crossing bundles can alter a diffusion measurement even when neither bundle has lost axons. The study examined a combined measure reflecting estimated fiber density and bundle cross sectional properties.

Reading ability was analyzed as a continuous variable rather than only as two diagnostic groups. This allowed the investigators to ask whether white matter properties tracked differences across a broad range of skills. The approach separates a relationship with current performance from the stronger claim that one anatomical measurement identifies the cause of every reading disability.



Meisler et al. (2022), Fig. 1.



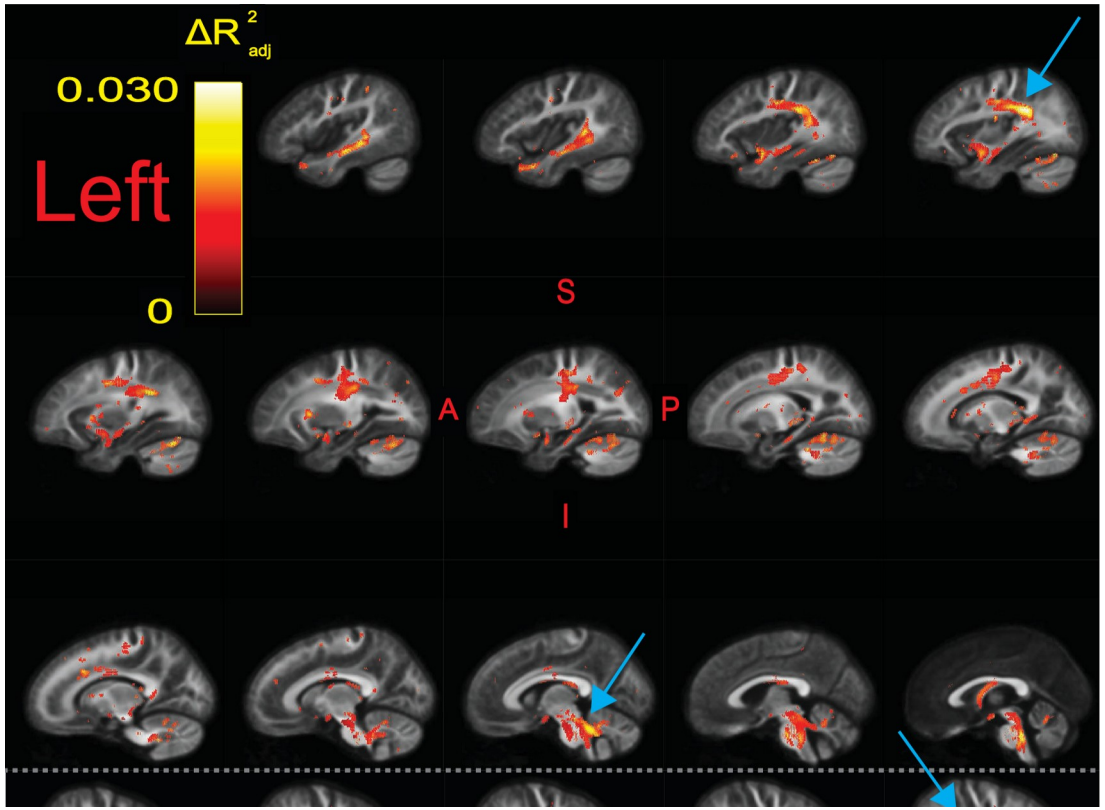
USC SPAN / Esling (2015), rtMRI: bod.

Reading skill relates to distributed fiber measures

The fiber specific reading study found positive relationships between reading skill and the combined fiber measure in multiple regions, especially left temporoparietal and cerebellar white matter. These pathways participate in distributed networks rather than one isolated visual word center. Better word reading was associated with differences in estimated tissue properties across that network.

The same analysis did not find a corresponding simple difference between the reading disability and comparison groups. A continuous association can therefore be present without producing a clean anatomical boundary between diagnostic categories. Individuals within each category vary, and the measured property need not change abruptly at the threshold used to define a reading difficulty.

The study's source images show the locations of fiber related associations, with the original color scale retained. The biological measurement remains an estimate derived from water diffusion and a tissue model. It describes structural properties associated with reading skill, while the behavioral tasks establish what the children could actually read accurately and quickly.



Meisler et al. (2022), Fig. 3.

Visual expertise stabilizes word recognition

Skilled readers recognize familiar written forms across differences in font, size, and case. This requires extracting stable combinations of features while remaining sensitive to distinctions that change a word. Ventral occipitotemporal cortex contributes to this learned visual expertise through interactions with language regions that supply pronunciation, meaning, and expectations about useful written patterns.

Writing systems reuse visual elements such as lines, curves, junctions, and repeated combinations. Reading experience makes those arrangements informative rather than turning the visual cortex into a completely separate sensory organ. A region can become more responsive to written words as literacy develops while continuing to participate in a wider network of visual recognition and language processing.

Reading also unfolds across eye movements, so the visual information available at one moment includes more than the fixated word. Nearby words fall outside the sharpest central portion of vision but can still influence subsequent processing. Experiments that track both gaze and brain activity test how much information is extracted before the eyes move directly onto the next word.

English	Arabic	Hindi	Mandarin
cars ears	زمان رمان	आज आजा	天 夫
	time/era pomegranate	today come	sky man

Carlson and Birkett (2022), Fig. 14.31, p. 485.



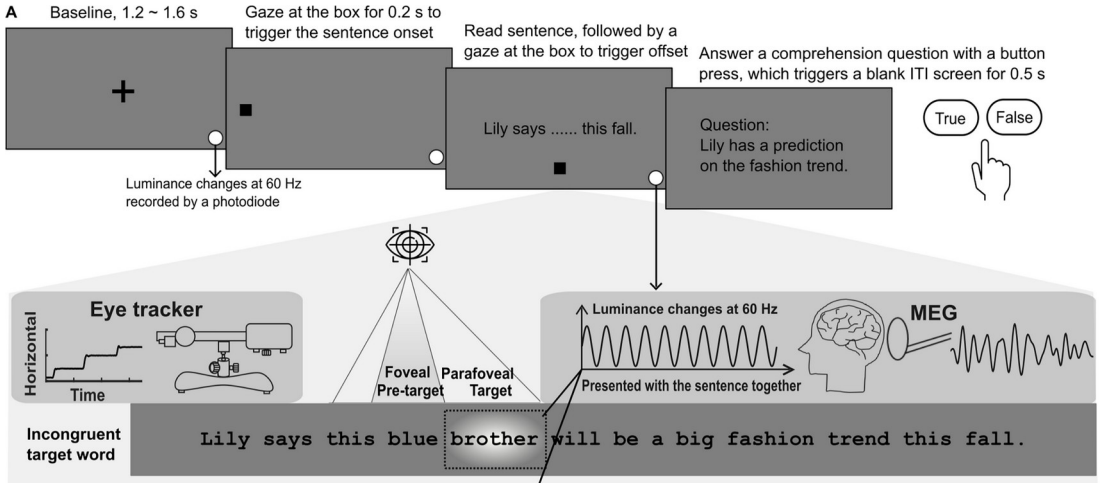
USC SPAN / Esling (2015), rtMRI: bawed.

Readers preview upcoming word information

A study combined eye tracking with magnetoencephalography while participants silently read sentences. A target word was either consistent or inconsistent with the preceding context. The investigators analyzed brain responses while the eyes were still on the word immediately before the target, asking whether the upcoming word already influenced processing from outside central fixation.

The target was flickered rapidly to tag its neural response without making the flicker consciously obvious. This technique allowed the researchers to separate activity related to the upcoming word from activity associated with the currently fixated word. The original paradigm figure shows when fixation, sentence presentation, and the relevant eye movements occurred during a trial.

The target words were selected so that the sentence did not make one exact word strongly predictable. The comparison therefore concerned whether a visible upcoming word fit the context, rather than simply whether the brain had guessed its identity in advance. Eye tracking established that the response being tested occurred before direct fixation of that target.



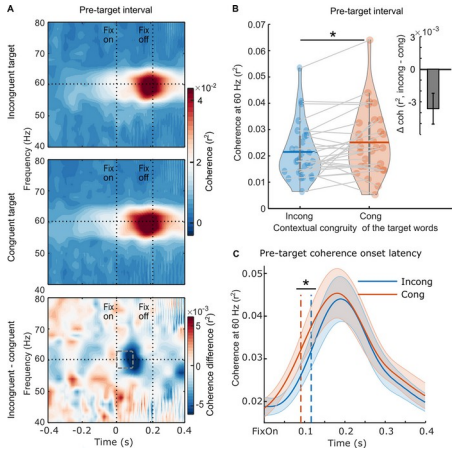
Pan et al. (2024), Fig. 1.

Parafoveal meaning interacts with context

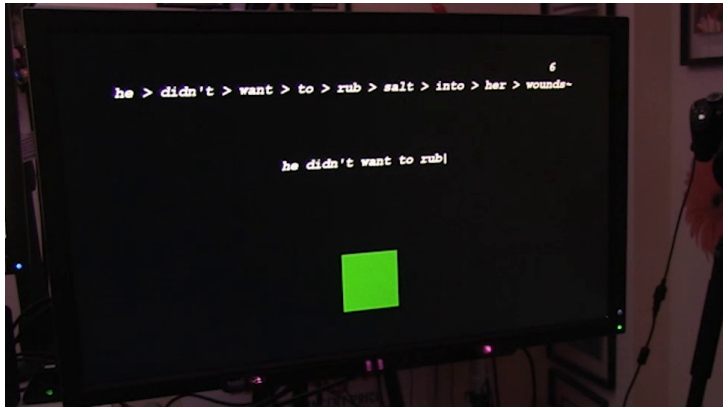
The natural reading study found a weaker tagged response for upcoming words that conflicted with the preceding sentence context. The difference was already detectable shortly after fixation of the preceding word. Parafoveal information therefore contributed to processing before the eyes landed directly on the target, including information relevant to whether its meaning fit the developing message.

The size of the response difference also related to individual reading speed. This association suggested that early processing of upcoming words contributes to the efficiency of natural reading. The neural signal did not require the participant to interrupt reading and report the parafoveal word, preserving the sequence of ordinary sentence comprehension more closely than a separate word identification task.

The result challenges an account in which only the currently fixated word can contribute semantic information. It does not imply equally detailed recognition across the entire visual field. The tested effect concerned a nearby target in a specific sentence position, with its neural response isolated by a frequency tag and its visibility verified through concurrent measurement of gaze.



Pan et al. (2024), Fig. 3.



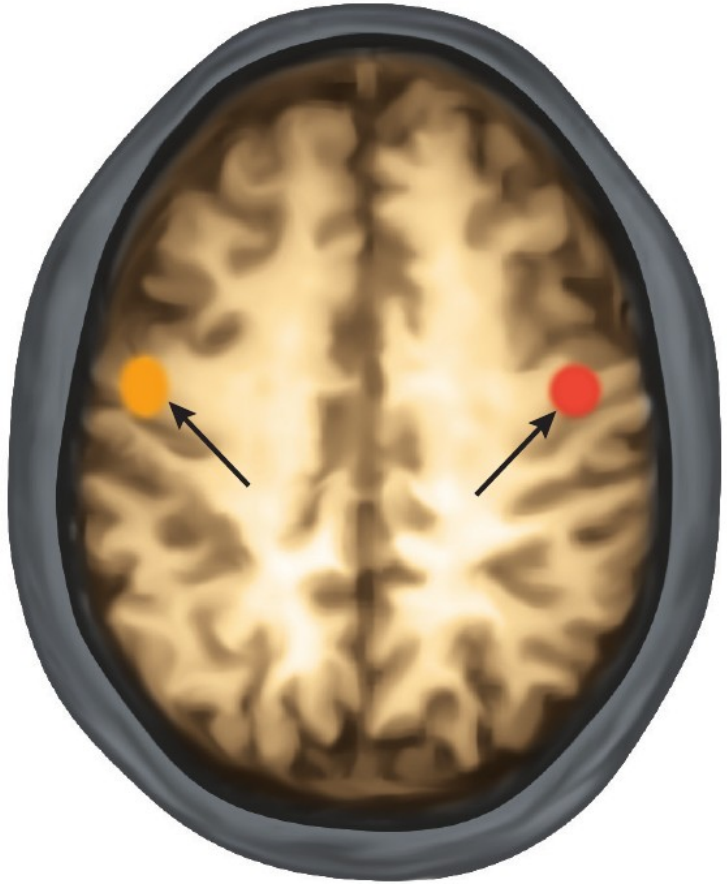
Willett et al. (2021), Video 1.

Writing requires spelling and motor organization

Writing transforms a message into an ordered sequence of visible marks. Spelling requires selecting the appropriate letters or characters, while motor planning organizes the movements that produce them. These components can fail separately. A person may know the intended spelling but struggle to form letters, or move the hand normally while making substantial spelling errors.

Phonological spelling converts a word's sounds into written units, which helps when the exact word has not been memorized. Orthographic knowledge supplies familiar spellings, including irregular forms that cannot be recovered reliably by sounding them out. Writing also uses temporary storage to preserve letter order while the hand or keyboard produces the sequence over time.

Premotor and parietal systems contribute to organizing learned writing movements. Typing and handwriting express similar linguistic content through different motor patterns, so damage can impair one more than the other. Neural interfaces can bypass a paralyzed limb by decoding intended movements, provided that the relevant motor representation remains accessible in the recorded brain signals.



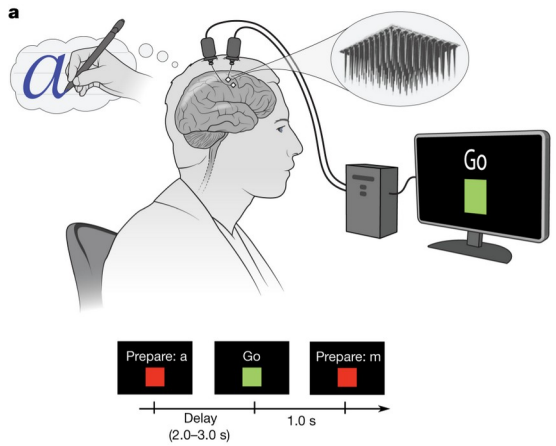
Carlson and Birkett (2022), Fig. 14.34, p. 488.

Attempted handwriting survives spinal injury

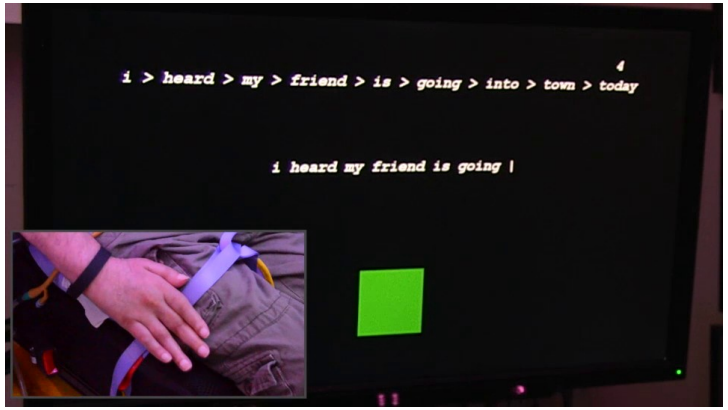
A handwriting brain computer interface study recorded from motor cortex in a participant with tetraplegia after a spinal cord injury. The participant attempted to write characters as if holding a pen. Microelectrode arrays measured activity related to those attempts, allowing a computer to estimate the intended character without requiring the hand to produce a readable mark.

The injury interrupted useful movement below the lesion but did not eliminate every cortical representation associated with learned handwriting. Different attempted characters produced distinguishable patterns across the recorded population. A trained recurrent neural network converted the changing signals into probabilities for characters and for the boundaries between them as the attempted writing unfolded.

The resulting text could be copied from a displayed sentence or generated to answer a question. These tasks require different levels of language formulation even when the same motor decoder produces the output. The interface restored an output route from preserved intention; it did not teach spelling to the participant or recover meanings lost through aphasia.



Willett et al. (2021), Fig. 1.



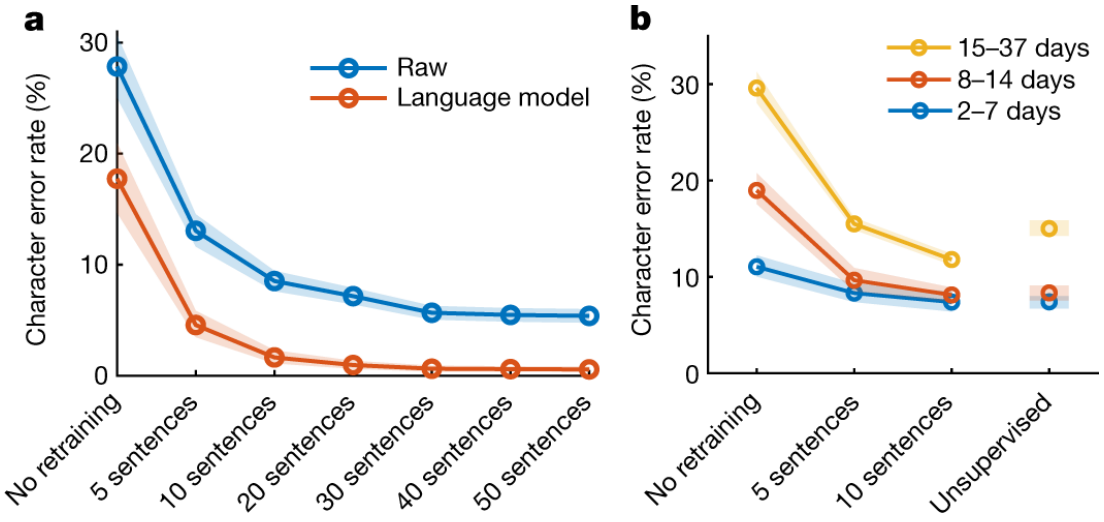
Willett et al. (2021), Video 2.

Character trajectories support text decoding

The handwriting interface exploited the changing neural patterns associated with drawing different characters. A character is a sequence of intended movements rather than one static hand position. Those trajectories can separate alternatives that would be difficult to distinguish from a small collection of sustained directional commands, making the temporal structure useful for identifying the intended symbol.

The study compared representations of attempted characters and evaluated decoding during real time use. A language model could help choose plausible character sequences after the neural decoder generated its estimates. The movement representation and the linguistic correction therefore supplied different sources of information: one from recorded cortical activity and another from regularities in written language.

Preserving that distinction matters when interpreting an interface demonstration. Correct final text can reflect both a strong neural signal and assistance from the language model. The participant's intended message still provides the target for evaluation, while the study separately examines how errors change when neural decoding and subsequent text correction are combined.



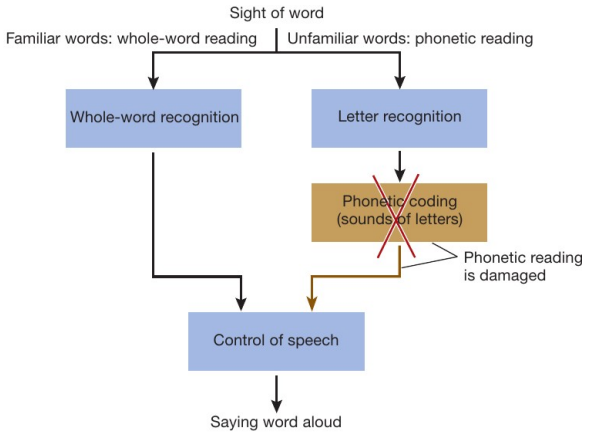
Willett et al. (2021), Fig. 3.

Phonological dysgraphia impairs sound based spelling

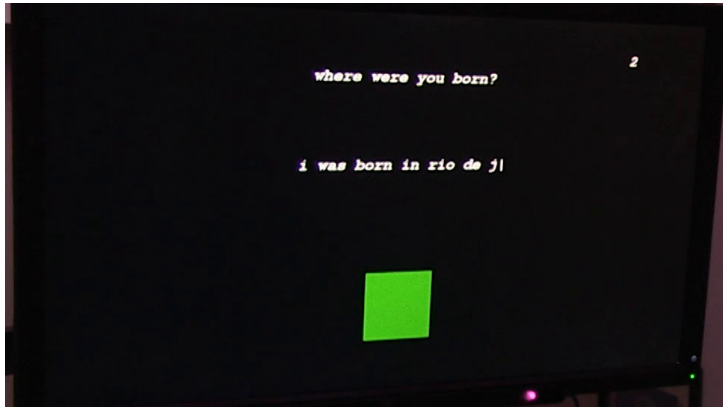
Phonological dysgraphia is an acquired difficulty converting speech sounds into written forms. A person may write familiar words from stored spelling knowledge while struggling with unfamiliar words or pronounceable nonwords. The motor ability to draw the letters can remain intact, showing that the problem concerns selecting the written sequence rather than simply moving a pen.

This pattern resembles phonological dyslexia on the input side of reading, but writing requires the transformation in the opposite direction. The person must hold a sound sequence, identify useful units, and select corresponding letters. Damage to frontal and related language systems can disrupt these operations while leaving some established orthographic memories available for frequently encountered words.

Testing copied writing alone would miss much of this impairment because copying supplies the visual sequence externally. Dictation, spontaneous writing, familiar words, and nonwords make different demands. Comparing them reveals whether the person can generate spelling from phonology, retrieve a familiar spelling, or reproduce a sequence that is already visible on the page.



Carlson and Birkett (2022), Fig. 14.28, p. 483.



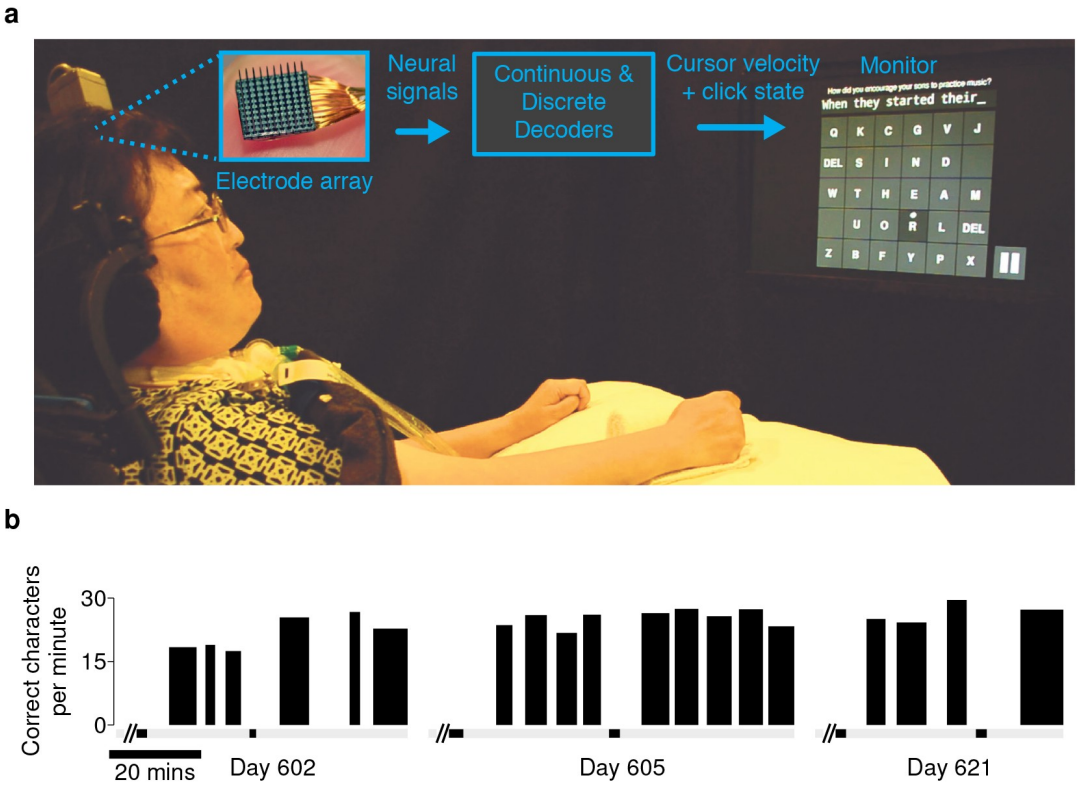
Willett et al. (2021), Video 3.

A cortical cursor enables written communication

An intracortical brain computer interface study enabled participants with paralysis to control a computer cursor and select characters on an onscreen keyboard. Arrays in motor cortex recorded activity that was translated into cursor movement and selection. This approach uses intended control signals to operate an external writing tool rather than directly decoding the linguistic meaning of a sentence.

In free response sessions, a participant formulated an answer to a question and then selected the required characters. Copy typing sessions instead supplied the intended text. Comparing the tasks distinguishes language generation from the speed and accuracy of selecting symbols, because a person can have a well formed answer while the interface remains slow to express it.

The system allowed different methods of character selection according to the participant's control abilities and preferences. Some used a click related neural signal, while another used dwelling on a target. These alternatives illustrate that restoring communication can involve adapting the output mechanism while leaving the person's own language and message selection intact.



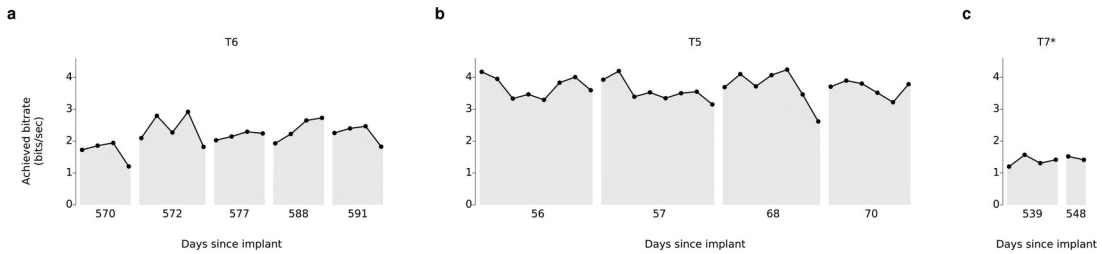
Pandarinath et al. (2017), Fig. 1.

Interface design influences symbol selection

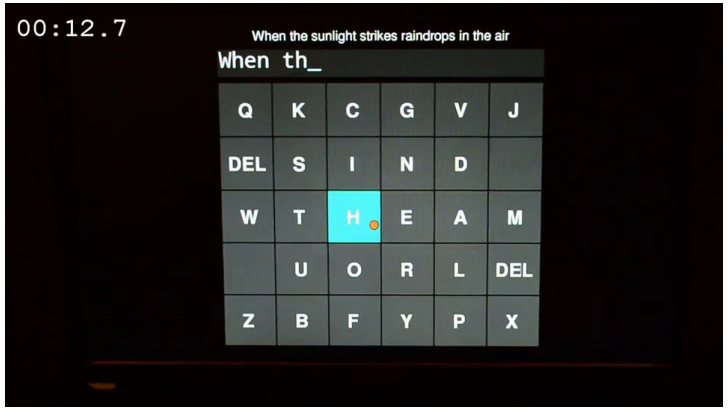
The cortical cursor study also measured performance on a grid of targets. A target selection task separates basic interface control from the particular words being typed. The computer specifies the desired location, and the participant moves and selects it. Errors and delays can then be attributed to the control task without requiring a judgment about spelling or sentence composition.

The study compared keyboard layouts and control procedures rather than assuming that one arrangement worked equally well for all participants. The position and size of targets influence how far the cursor must move and how precisely it must stop. A neural decoder therefore interacts with the geometry of the external interface to determine useful communication performance.

The original grid results quantify throughput under specific task conditions. They should not be read as a direct measurement of how quickly the brain generates ideas. Formulating an original message, correcting a spelling error, and selecting an externally prompted square involve different demands, even when the same recording arrays and cursor control system are used.



Pandarinath et al. (2017), Fig. 3.



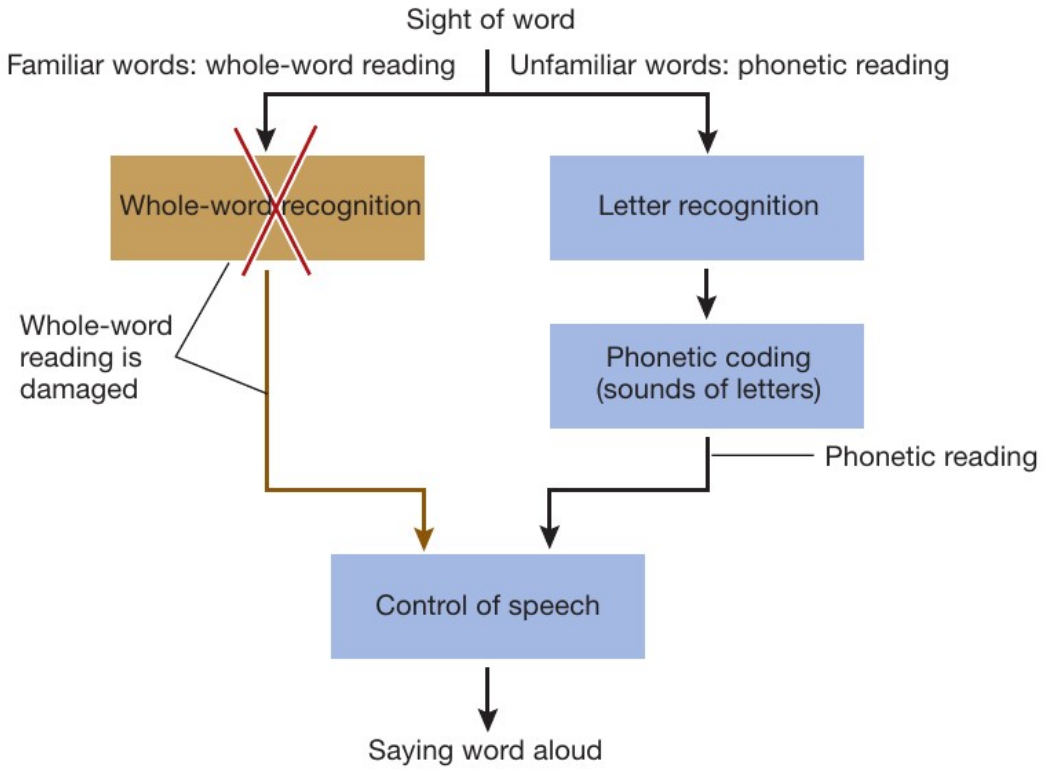
Pandarinath et al. (2017), media2.

Orthographic dysgraphia impairs stored spellings

Orthographic dysgraphia is an acquired impairment in retrieving established written word forms. A person may still convert sounds into plausible letters but produce spellings such as bizzy for busy. The response preserves much of the pronunciation while failing to recover the conventional orthographic sequence, revealing a different problem from impaired formation of the letter shapes.

The contrast with phonological dysgraphia demonstrates that spelling draws on more than one source of information. Familiar irregular words place strong demands on stored orthography, whereas nonwords require sound based conversion. A person with damage to one system can sometimes compensate with the other, making the pattern of correct and incorrect responses more informative than an overall writing score.

Inferior temporal language related regions contribute to stored orthographic knowledge, while frontal and parietal systems also support selection, order, and output. The same intended message may remain available in speech when its conventional written form is difficult to retrieve. Restoring a motor output route cannot by itself replace the missing spelling knowledge needed to compose that message correctly.



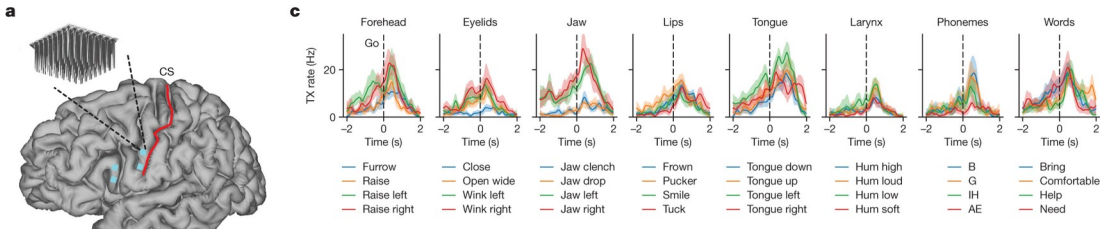
Carlson and Birkett (2022), Fig. 14.27, p. 482.

Attempted speech remains neurally represented

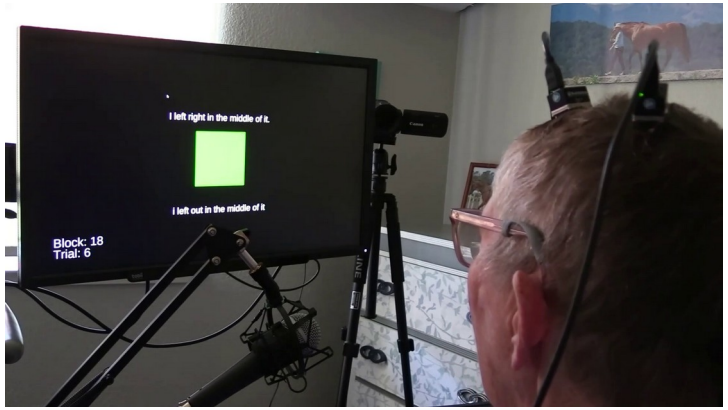
A speech neuroprosthesis study recorded cortical activity from a participant with amyotrophic lateral sclerosis whose speech had become unintelligible. Small electrode arrays sampled activity in regions including ventral premotor cortex. The participant attempted words, phonemes, and orofacial movements, allowing the investigators to test whether useful speech related representations persisted despite the loss of effective spoken communication.

Ventral premotor signals distinguished attempted speech and movements of the speech articulators. These responses provided information that could be used for decoding even when the sound produced by the vocal tract did not communicate the message reliably. The neurological distinction is between a damaged output capability and the continued availability of cortical information about intended articulatory behavior.

The study also sampled another frontal region, and the usefulness of recorded signals differed by location. Implanting an array in any language associated area therefore does not guarantee the same decoding performance. A successful interface depends on where activity is recorded, which signals remain informative, and how those signals are translated into a sequence of possible speech units.



Willett et al. (2023), Fig. 1.



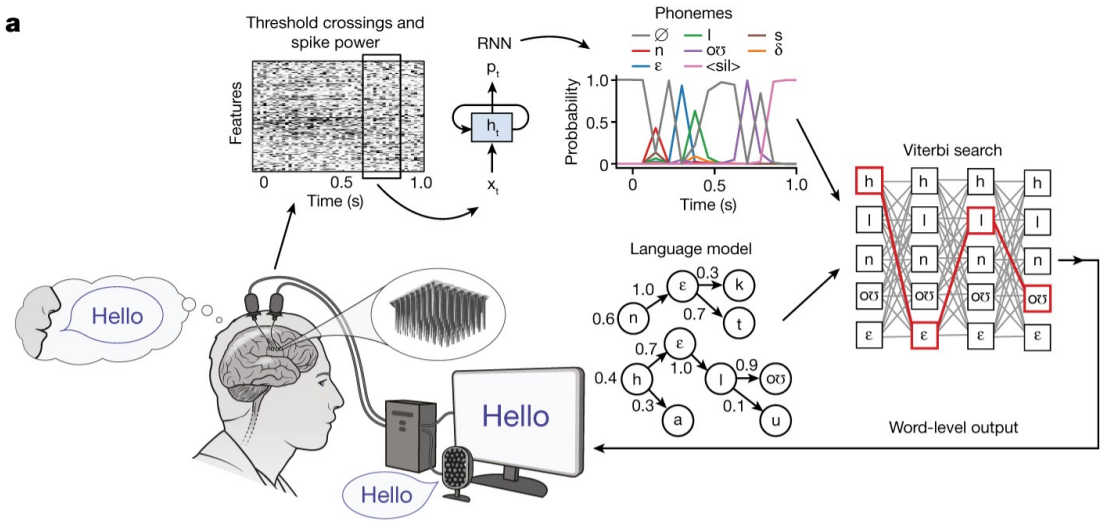
Willett et al. (2023), Video 1.

A decoder transforms neural signals into phonemes

The speech neuroprosthesis processed neural activity as a changing sequence rather than classifying one whole sentence at once. A recurrent neural network estimated probabilities for phonemes, the sound units that distinguish words. A language model then used those estimates to identify plausible word sequences, allowing the system to handle a much larger vocabulary than a small fixed list of commands.

The participant used the system in real time while attempting sentences. The published demonstrations include attempts with vocalization and a condition in which words were mouthed silently. Decoding did not require the external microphone to recognize intelligible speech; it used cortical signals associated with the attempted production of the message.

The final displayed sentence results from both the neural estimates and the language model's constraints. An error can arise when the recorded pattern is ambiguous or when linguistic regularities favor the wrong word sequence. Evaluating new sentences tests whether the system generalizes beyond memorized training examples while retaining the participant's intended content as the standard for accuracy.



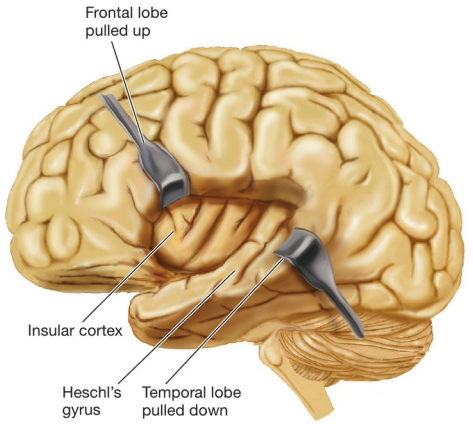
Willett et al. (2023), Fig. 2.

Paralysis and aphasia affect different stages

Paralysis can prevent the movements needed for speech or writing while leaving language comprehension and message formulation substantially preserved. Aphasia instead affects language operations themselves, such as retrieving words or interpreting sentences. The conditions can coexist, but they are not interchangeable explanations for why a person cannot produce a conventional spoken or written response.

A motor neuroprosthesis attempts to provide an alternative output route from preserved neural representations. Its success therefore depends partly on which representations remain available. A decoder trained on attempted speech movements does not automatically recover a concept that the person can no longer identify or a word whose meaning has been lost through progressive temporal degeneration.

Assessment must also consider sensory access and the person's preferred communication form. Written text, synthesized speech, signing, and symbol selection place different demands on perception and language. Matching an interface to retained abilities begins with identifying the disrupted biological stage rather than assuming that every communication difficulty can be solved by bypassing the same motor pathway.



Carlson and Birkett (2022), Fig. 14.9, p. 467.



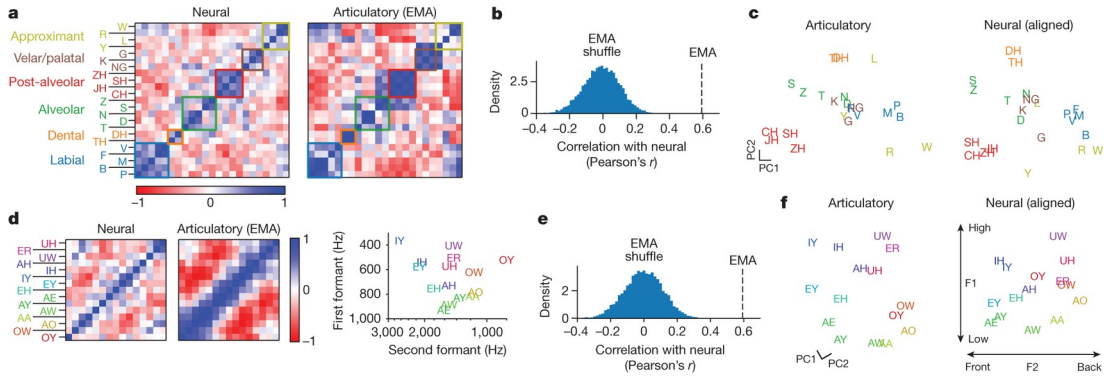
Willett et al. (2023), Video 2.

Phoneme geometry reflects articulation

The speech neuroprosthesis study compared the similarity of neural patterns for different consonants with the similarity of their articulatory movements. Consonants produced at related places in the vocal tract showed corresponding organization in the recorded neural representation. This pattern indicated that useful aspects of articulatory structure remained present despite severe impairment of spoken output.

The investigators also examined vowels and the organization of attempted speech units in lower dimensional spaces. These analyses describe relationships across population patterns rather than a one electrode one sound code. A particular electrode can contribute to several distinctions, while the collective activity provides a structured basis for estimating which phoneme the participant attempted.

The comparison used articulatory information from people able to speak normally as a reference, rather than measuring every hidden movement inside the participant's vocal tract during each attempt. Similar representational geometry supports a relationship to speech production structure. It does not establish that all cortical speech representations remain unchanged after every disease or duration of paralysis.



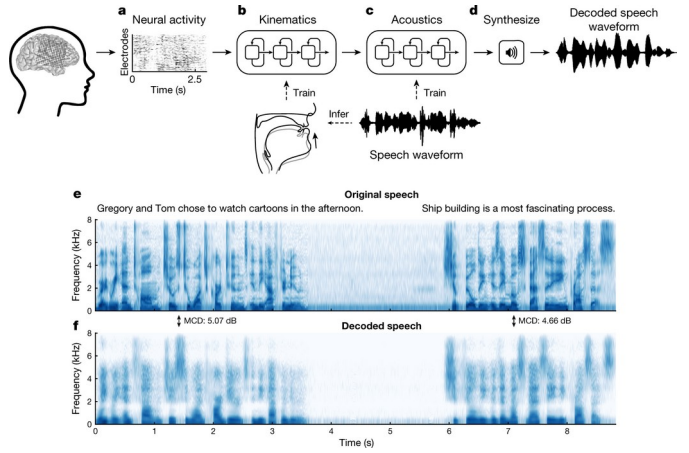
Willett et al. (2023), Fig. 3.

Speech synthesis can use movement estimates

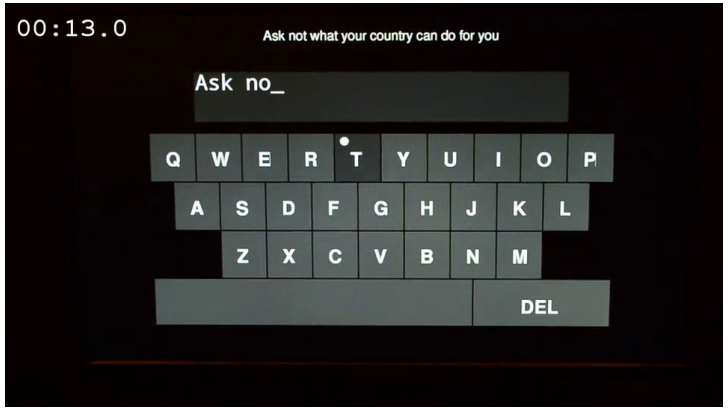
An earlier speech synthesis study recorded cortical activity from participants who could speak while undergoing clinical monitoring. The researchers developed a two stage decoder: neural activity first estimated articulatory movements, and those movement estimates then generated acoustic speech features. The approach used the physical organization of speech production as an intermediate representation between brain signals and sound.

The articulatory representation described coordinated movements of structures such as the tongue, lips, and jaw. These movements shape the acoustic signal, so related gestures can be shared across many words and sentences. A decoder that learns those relationships can reuse information about speech mechanics rather than learning an independent neural pattern for every possible sentence.

The published demonstration includes synthesized output and the original recorded speech, allowing the reconstructed sound to be compared with the actual utterance. This was a synthesis experiment using neural recordings from speaking participants. Its preparation differs from an implant tested for everyday communication in a person unable to produce intelligible speech, even though both contribute to the development of speech interfaces.



Anumanchipalli et al. (2019), Fig. 1.



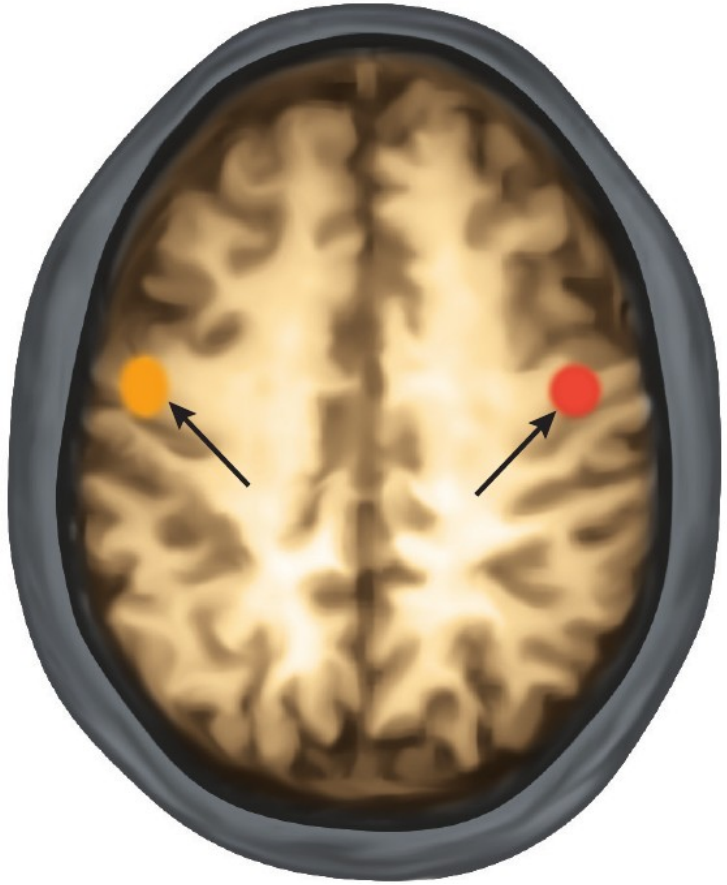
Pandarinath et al. (2017), media3.

Communication support exploits preserved routes

Alternative communication methods exploit abilities that remain available when a customary output is difficult. A person with preserved language can compose a message through eye movements, switch control, a cursor, or a neural interface. Each method translates an intentional choice into symbols, but the speed and effort required depend on the motor and sensory systems used for control.

Language impairments require a different analysis because the intended words or meanings may themselves be difficult to retrieve. Visual choices, context, partner support, and reduced sentence complexity can sometimes improve access, depending on the person's profile. The relevant goal is an effective exchange of the person's message rather than producing one particular form of speech at all costs.

The success of a support system is therefore assessed through actual communication tasks. Copying a displayed sentence, answering an original question, and participating in conversation make different demands on formulation, timing, and error correction. A device demonstration should specify which task was performed and which abilities the user supplied, rather than treating every accurate output as evidence of restored language processing.



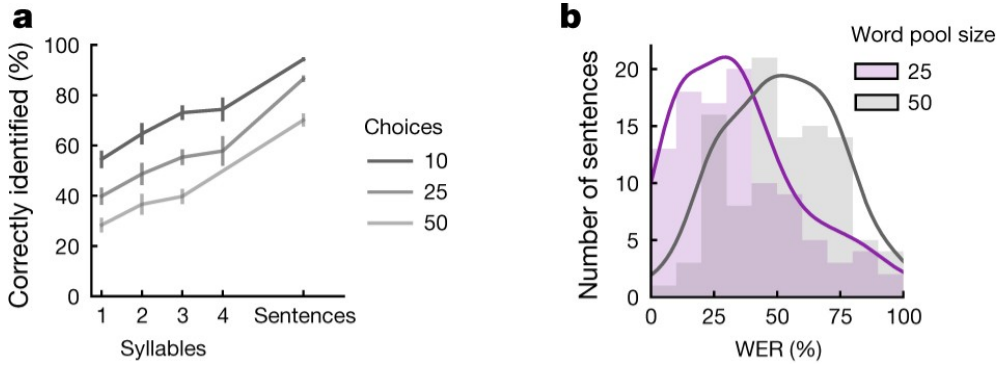
Carlson and Birkett (2022), Fig. 14.34, p. 488.

Articulatory decoding generalizes across material

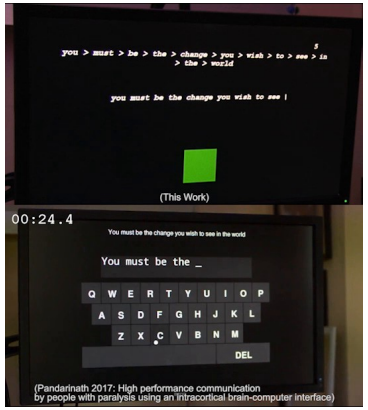
The neural speech synthesis study evaluated how well estimated articulatory representations could produce speech beyond the examples used to train the model. The intermediate representation reduced the problem to recurring movement patterns that contribute to many utterances. The original figure compares aspects of generated speech with the corresponding recorded material, making the decoder's target explicit.

Listeners could evaluate the intelligibility of synthesized output, while acoustic analyses quantified similarities to the original signal. These are related but different outcomes: matching some spectrotemporal features does not guarantee that every word is understandable. The experiment therefore assessed reconstructed speech through measures tied to the sound and the message rather than relying only on a model's internal error.

The decoder's articulatory stage was a learned estimate, not a direct camera recording of the vocal tract during every neural trial. Training data supplied relationships among neural activity, movement, and acoustics. Keeping these sources distinct explains how a physiologically organized model can help synthesize speech without implying that the computer directly reads a person's abstract thoughts.



Anumanchipalli et al. (2019), Fig. 2.



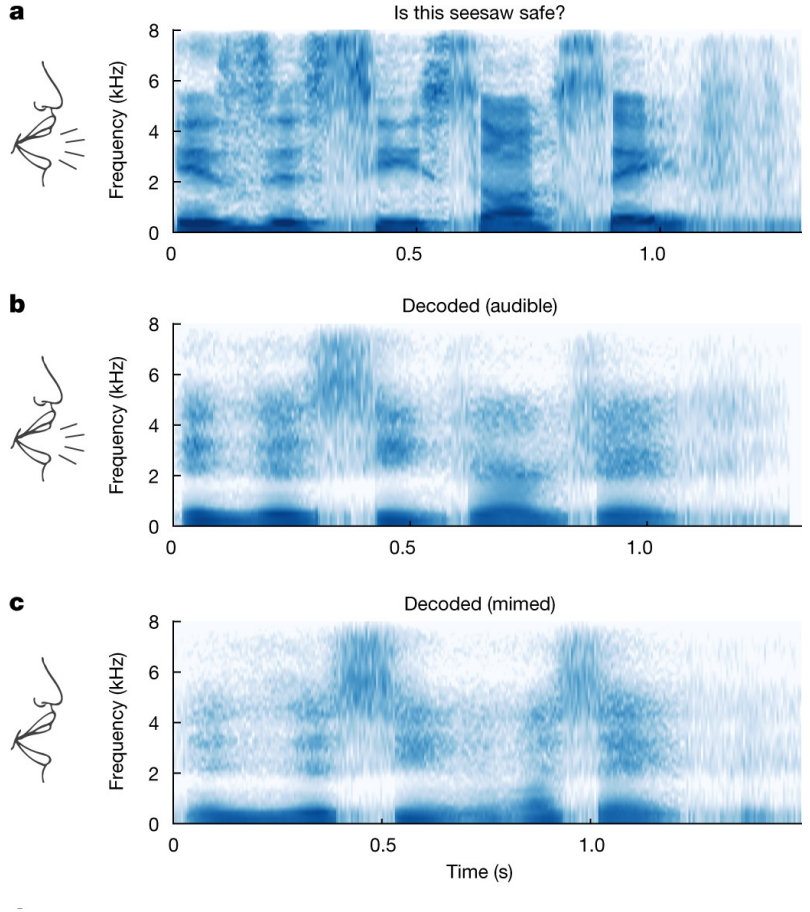
Willett et al. (2021), Video 4.

Silent miming differs from unrestricted inner speech

The speech synthesis work also examined a condition in which a participant silently mimed speech. The person attempted articulatory behavior without producing the same ordinary voiced signal used during spoken trials. The demonstration tested whether the neural representation could still support reconstruction when the usual acoustic output was absent or substantially reduced.

Silent miming is not the same as unrestricted inner speech or a private thought. It is an instructed task involving an attempted speech action with a known experimental target. The distinction matters because decoding performance during that task cannot establish that the system can recover any sentence a person happens to imagine without an intention to communicate it.

The result extended the interface concept toward conditions in which acoustic speech is unavailable, while leaving the training and task requirements visible. Together with later attempted speech studies, it shows how the target can shift from reconstructing spoken sentences to providing a controllable output route. The participant's deliberate communicative action remains central to interpreting what the decoder actually measures.



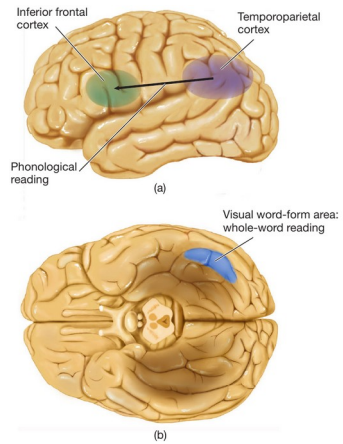
Anumanchipalli et al. (2019), Fig. 3.

Spelling pronunciation and meaning can dissociate

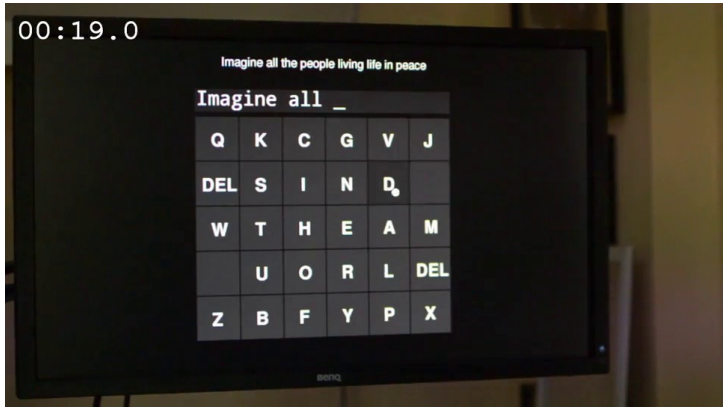
A written word can be recognized visually, pronounced, understood, and produced through partially separable operations. Acquired reading and writing disorders reveal these distinctions when one transformation fails. A correct spoken rendition does not guarantee comprehension, and a meaningful spoken answer does not guarantee that the person can retrieve its conventional written spelling.

Reading tasks therefore vary the kind of information that can support success. Familiar irregular words test stored word knowledge, while nonwords test conversion between written units and sounds. Writing tasks reverse some of those demands and additionally require preserving symbol order through time. The pattern across tasks identifies the affected process more clearly than one combined accuracy score.

Neural interfaces add another dissociation by replacing the motor route used to express a written message. They can preserve the distinction between language formulation and physical output because the participant still chooses the content. A system that decodes attempted handwriting bypasses a movement limitation; it does not make acquired dyslexia or dysgraphia disappear by generating a correct looking string on a screen.



Carlson and Birkett (2022), Fig. 14.30, p. 484.



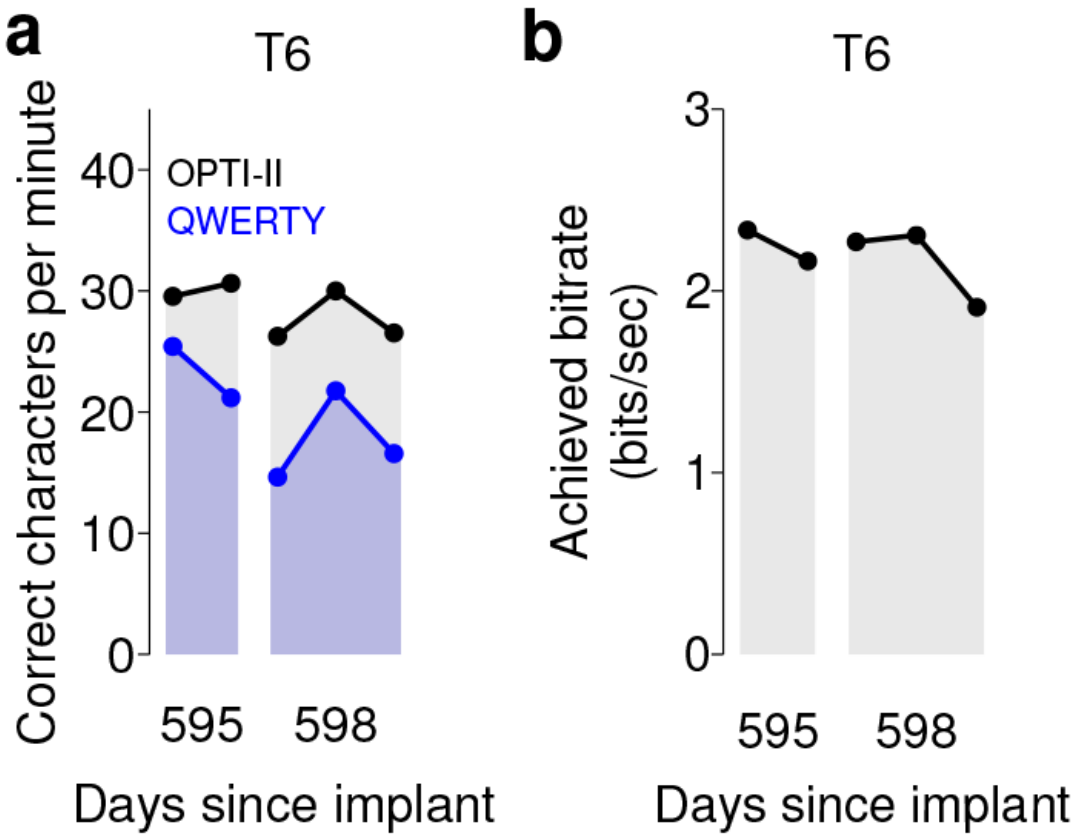
Pandarinath et al. (2017), media4.

Cursor control persists without overt movement

The cortical cursor study tested whether performance depended on residual physical movements in a participant who retained some hand function. The participant completed control tasks while instructed to suppress those movements. This comparison examined whether useful interface signals were available without relying on the hand to move a conventional mouse or activate an external selection switch.

Performance remained possible under the movement suppression condition. The result supported the interpretation that the neural interface provided control through recorded cortical activity rather than merely detecting an ordinary external movement. It also distinguished that participant's retained motor abilities from those of other participants who had no useful limb movement during interface use.

The test does not imply that motor cortical activity becomes unrelated to movement intention. Intended actions are precisely the information used to operate the cursor. It shows that an overt limb displacement is not always required for a useful cortical control signal, allowing a writing tool to be operated through an alternative pathway when the usual muscular route is unavailable.



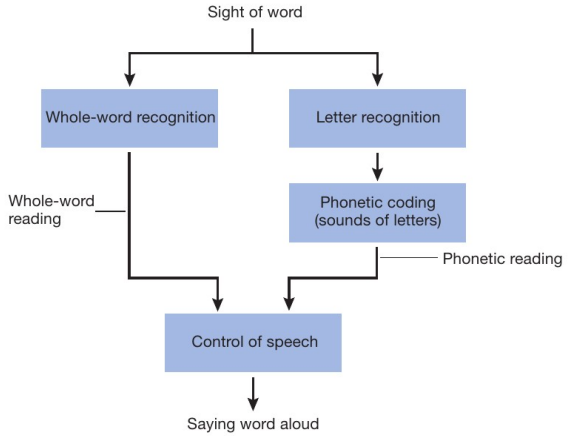
Pandarinath et al. (2017), Fig. 4.

Task comparisons locate communication failures

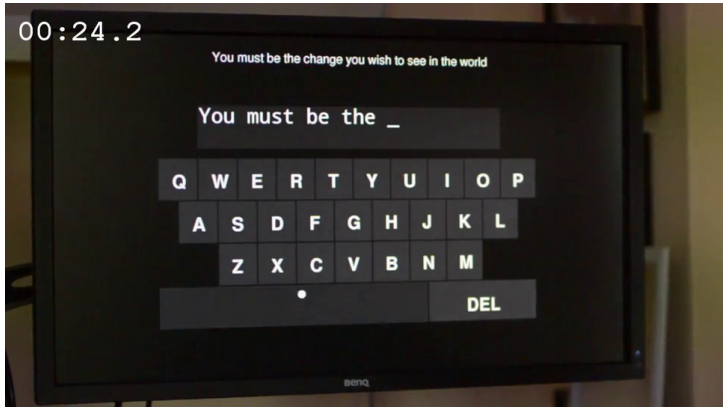
A person's communication profile emerges from comparisons across tasks and modalities. Hearing a word, recognizing a written word, understanding a sign, naming a picture, repeating a nonword, and writing to dictation each require overlapping but different operations. A selective difficulty identifies a stage that cannot be inferred from the absence of fluent speech alone.

For example, preserved comprehension with impaired repetition suggests a different disturbance from fluent repetition without comprehension. Preserved writing with impaired reading points toward disrupted visual access, while intact spelling knowledge with an unusable hand points toward an output limitation. These contrasts connect a behavioral finding to the pathway that must transform information to support that response.

Anatomical and physiological measurements refine that account by identifying damaged tissue, surviving networks, and the timing of neural activity. They do not replace the behavioral comparison. A brain image shows structure or a measured signal; the communication task establishes which message the person could understand or express through the tested sensory and motor route.



Carlson and Birkett (2022), Fig. 14.26, p. 481.



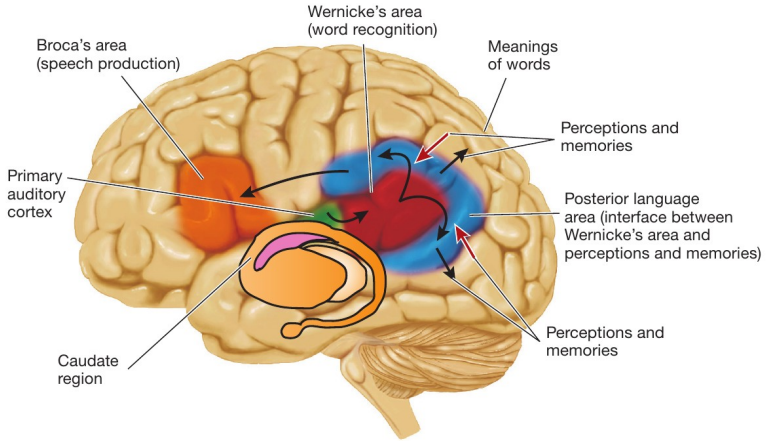
Pandarinath et al. (2017), media5.

Communication requires coordinated neural systems

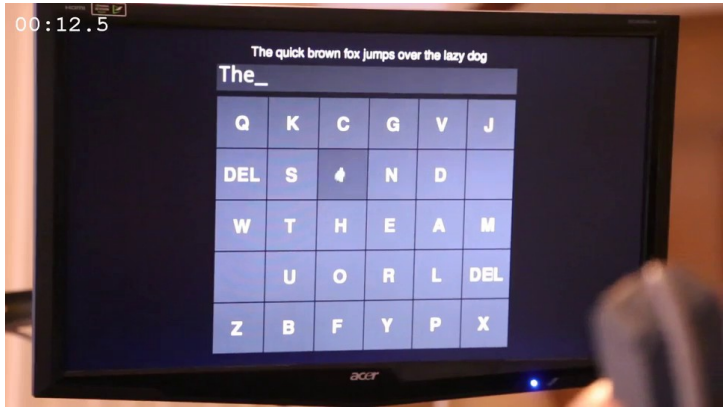
Communication transforms sensory patterns into meaningful representations and selected messages into organized actions. Different pathways contribute to recognizing word forms, interpreting concepts, assembling sentence structure, and controlling speech or writing. Their interactions allow a listener to become a speaker and a reader to become a writer without requiring every component to perform the same operation.

Lesions and degenerative disease reveal separable components because understanding, naming, repetition, reading, and writing can change independently. Direct recordings add timing and representational detail, showing that the same broad cortical region can carry different information during listening and speaking. Developmental and cross linguistic studies further show how experience shapes the networks used for these transformations.

Restoring communication begins with the abilities that remain available. Accessible language, targeted support, and alternative output routes address different biological problems. A speech decoder or handwriting interface is most interpretable when the preparation, intended action, and measured output are specified, preserving the distinction between expressing a message and understanding the language from which that message was formed.



Carlson and Birkett (2022), Fig. 14.3, p. 460.



Pandarinath et al. (2017), media6.

Key takeaways

Language uses interacting distributed systems. Sensory word analysis, conceptual knowledge, sentence organization, and motor output can change independently; hemispheric dominance does not make language a single-center function.

Speech responses depend on the operation being performed. Direct recordings distinguish preparation, articulation, sensory feedback, and listening; activation in motor cortex alone does not identify an articulatory code.

Aphasia requires component-level assessment. Compare comprehension, naming, repetition, grammar, reading, writing, and signed language to identify which transformations remain available.

Experience shapes language and literacy networks. Early accessible language, linguistic rhythm, and writing-system experience influence neural representations; deafness and language deprivation are distinct conditions.

Reading and writing combine several routes. Stored orthography, sound conversion, semantics, and motor sequence knowledge explain dissociations among acquired reading and spelling disorders.

Neuroprostheses bypass particular output limitations. Speech, handwriting, and cursor decoders use preserved intentional signals; restoring a motor route differs from repairing lost word meanings or grammatical knowledge.

Carlson & Birkett (2022); primary studies in notes